

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



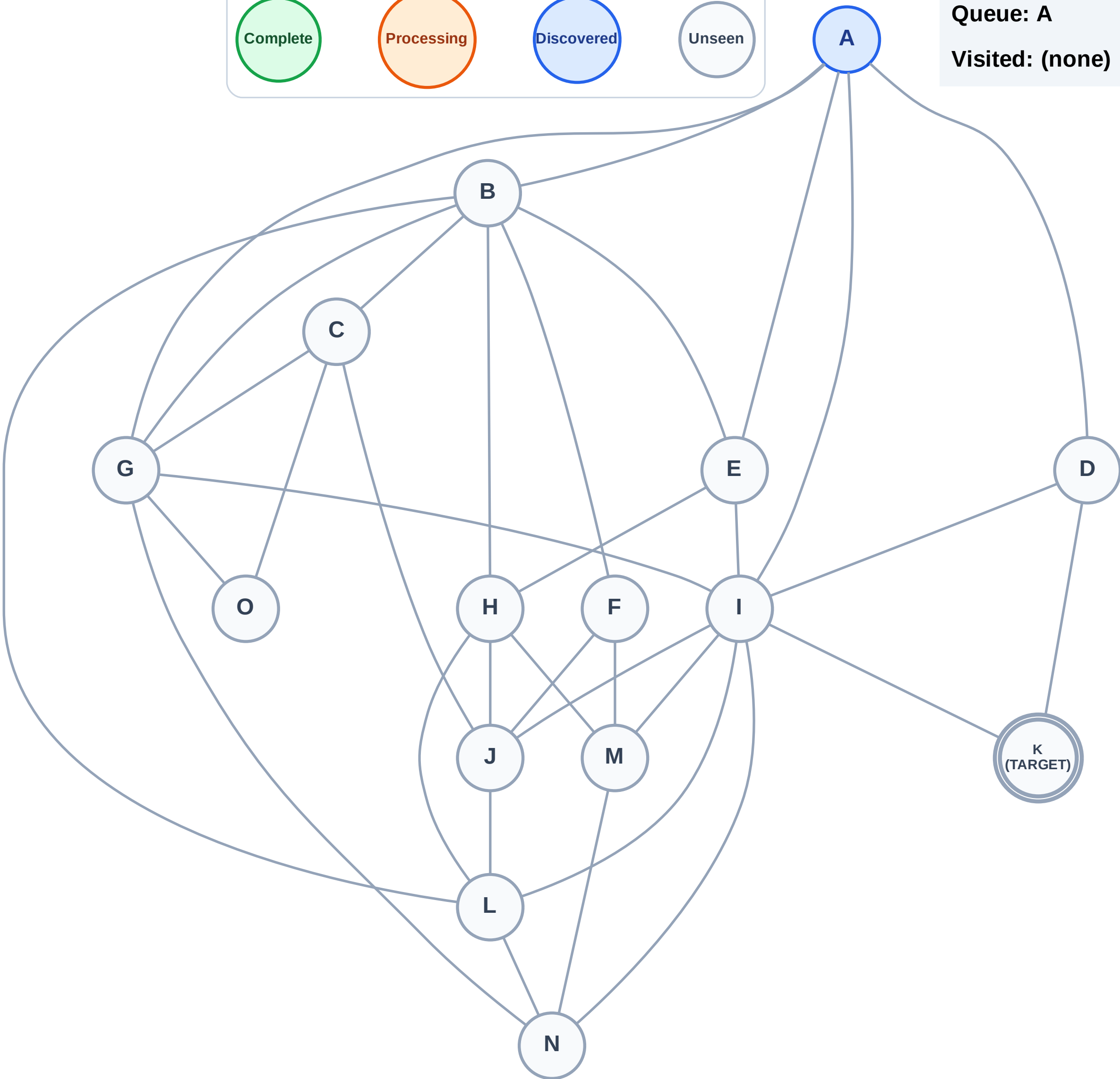
Discovered



Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



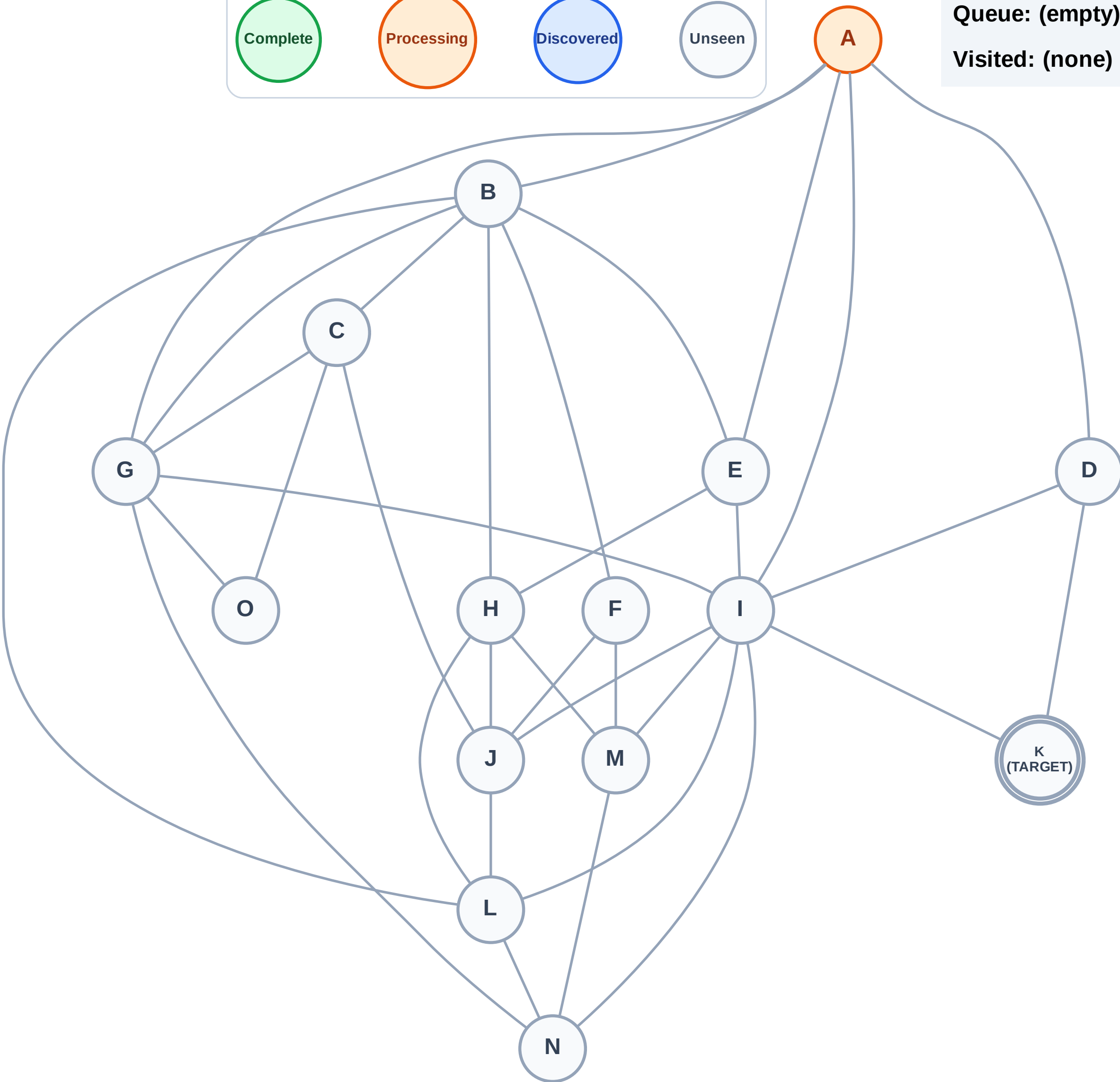
Discovered



Unseen

Queue: (empty)

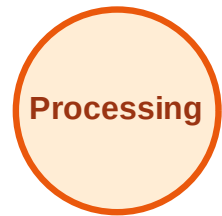
Visited: (none)



BFS Traversal

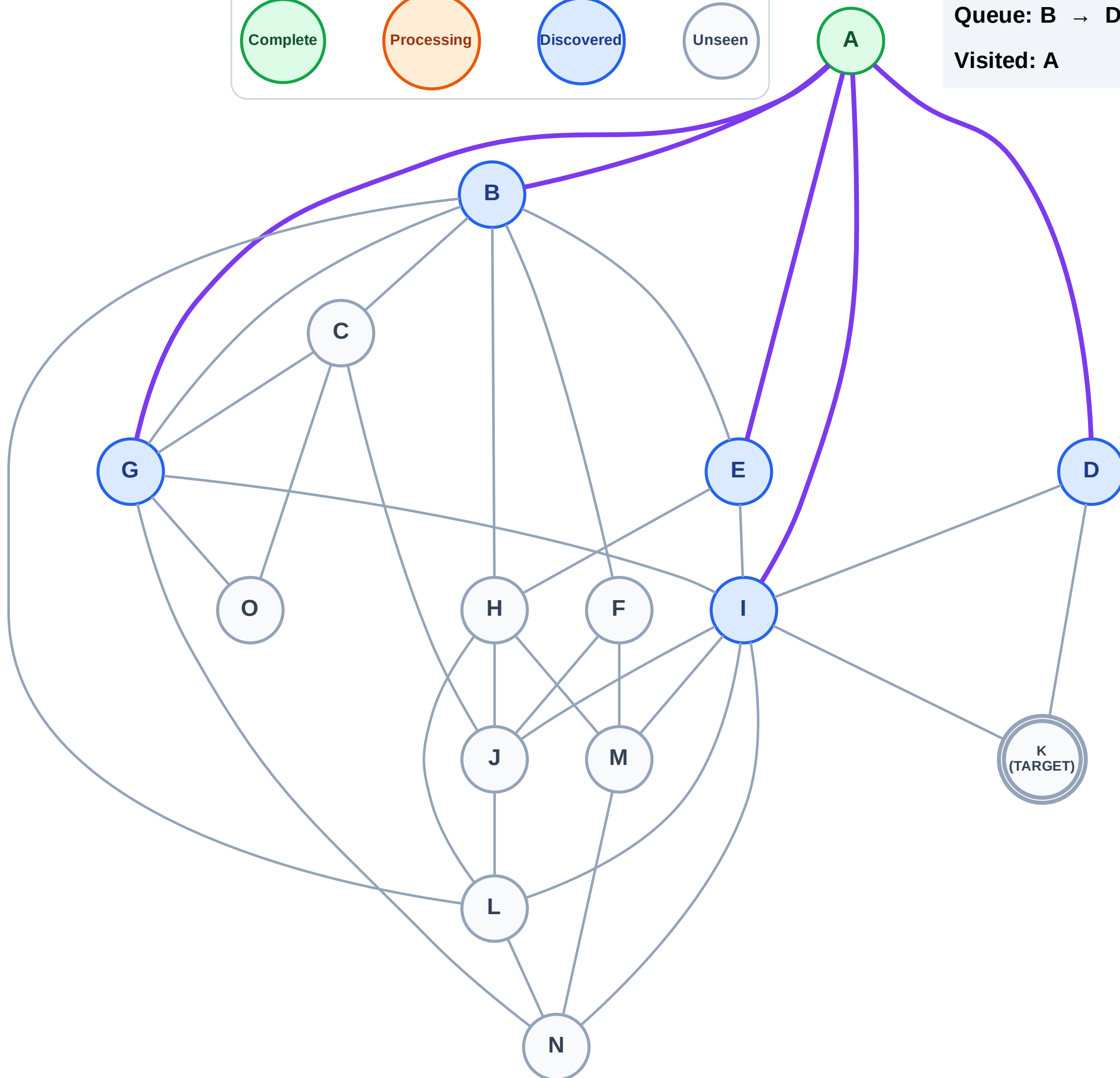
Step 3: Discover B, D, E, G, I from A

Legend



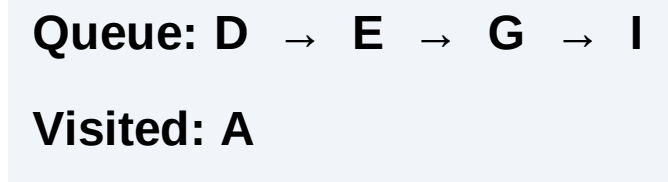
Queue: B → D → E → G → I

Visited: A



Step 4: Process node B

Step 4: Process node B



BFS Traversal

Step 5: Discover C, F, H, L from B

Legend



Complete



Processing



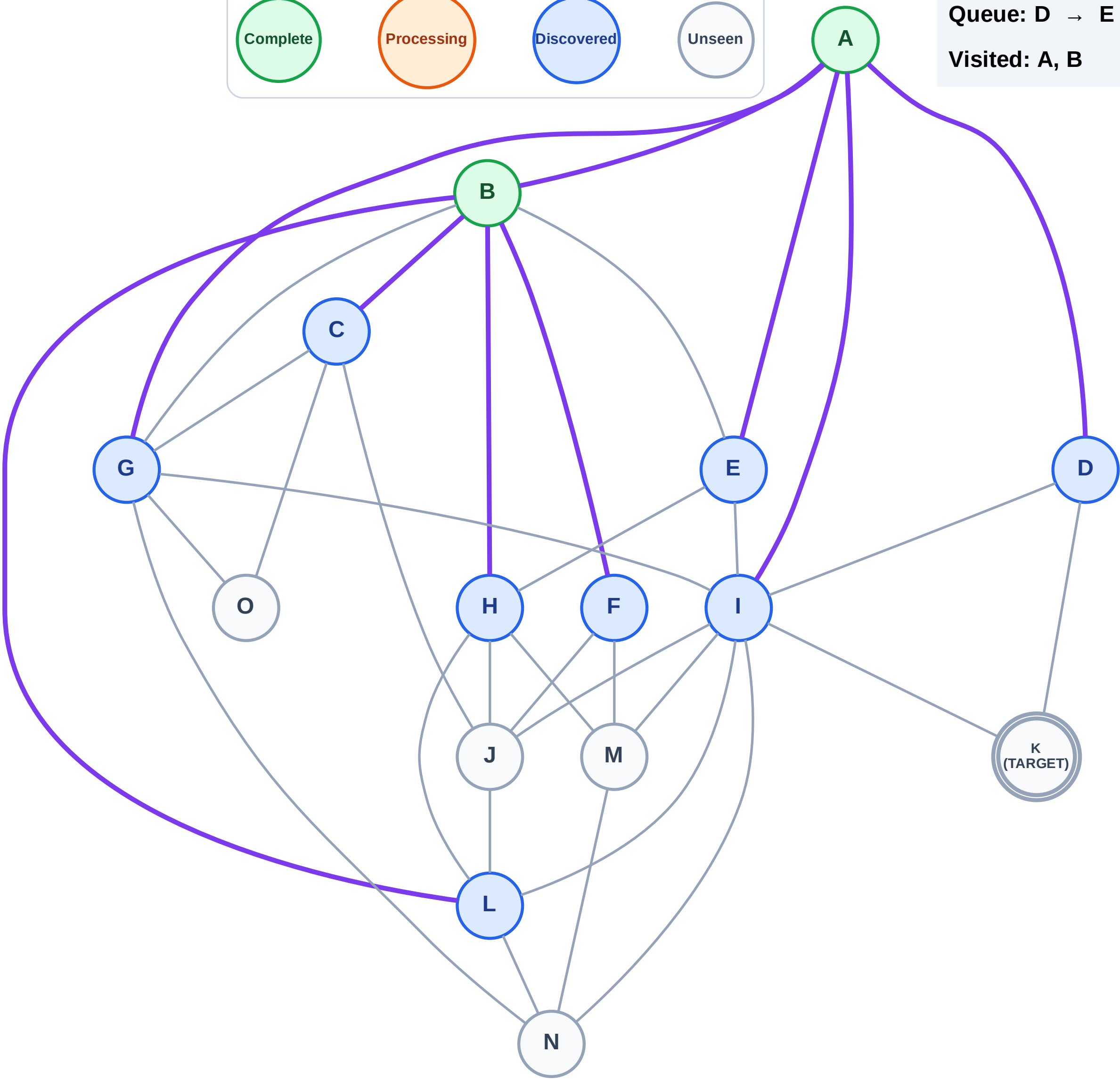
Discovered



Unseen

Queue: D → E → G → I → C → F → H → L

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

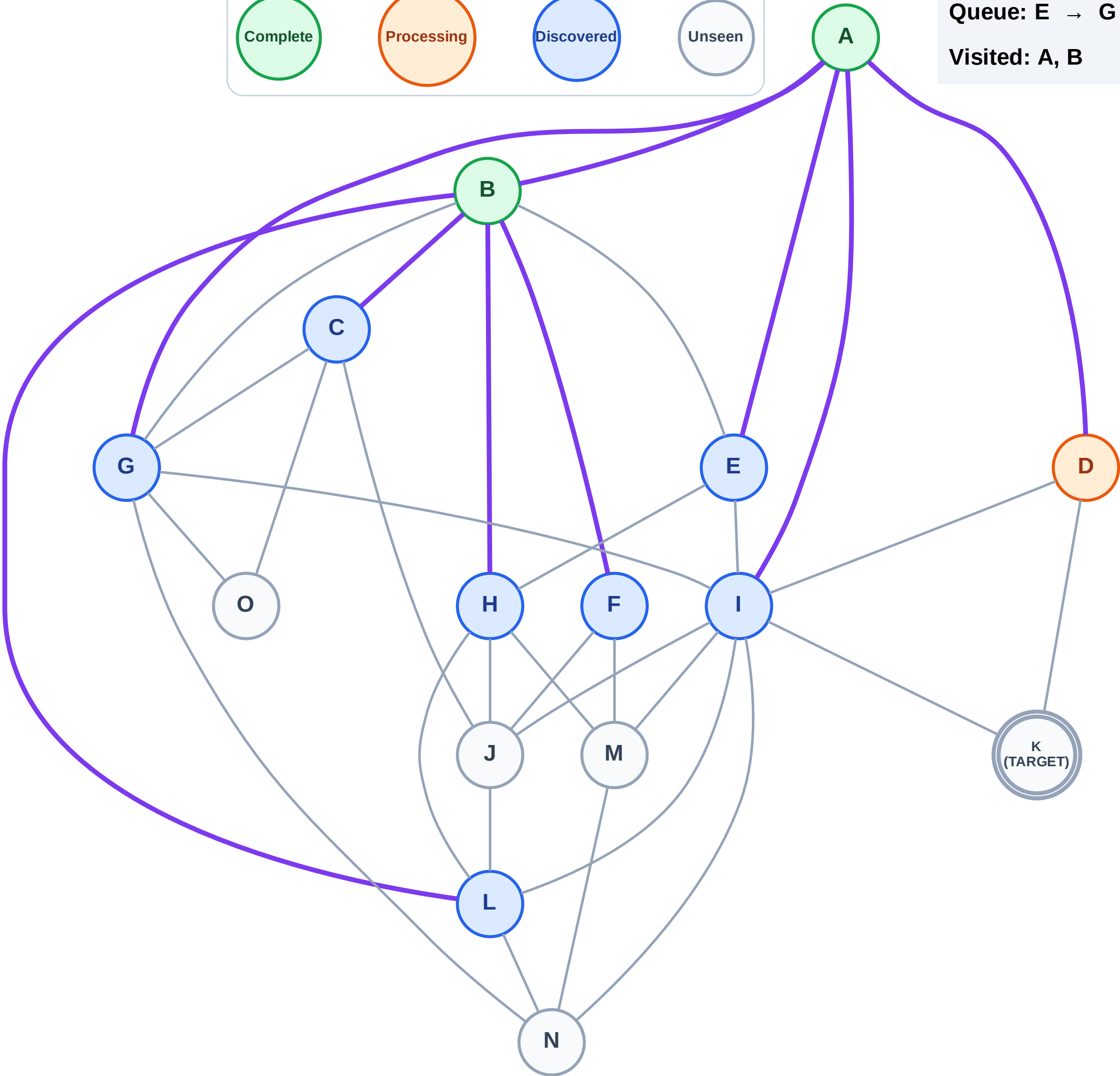
Processing

Discovered

Unseen

Queue: E → G → I → C → F → H → L

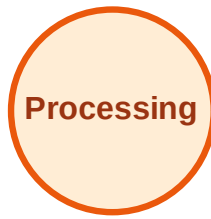
Visited: A, B



BFS Traversal

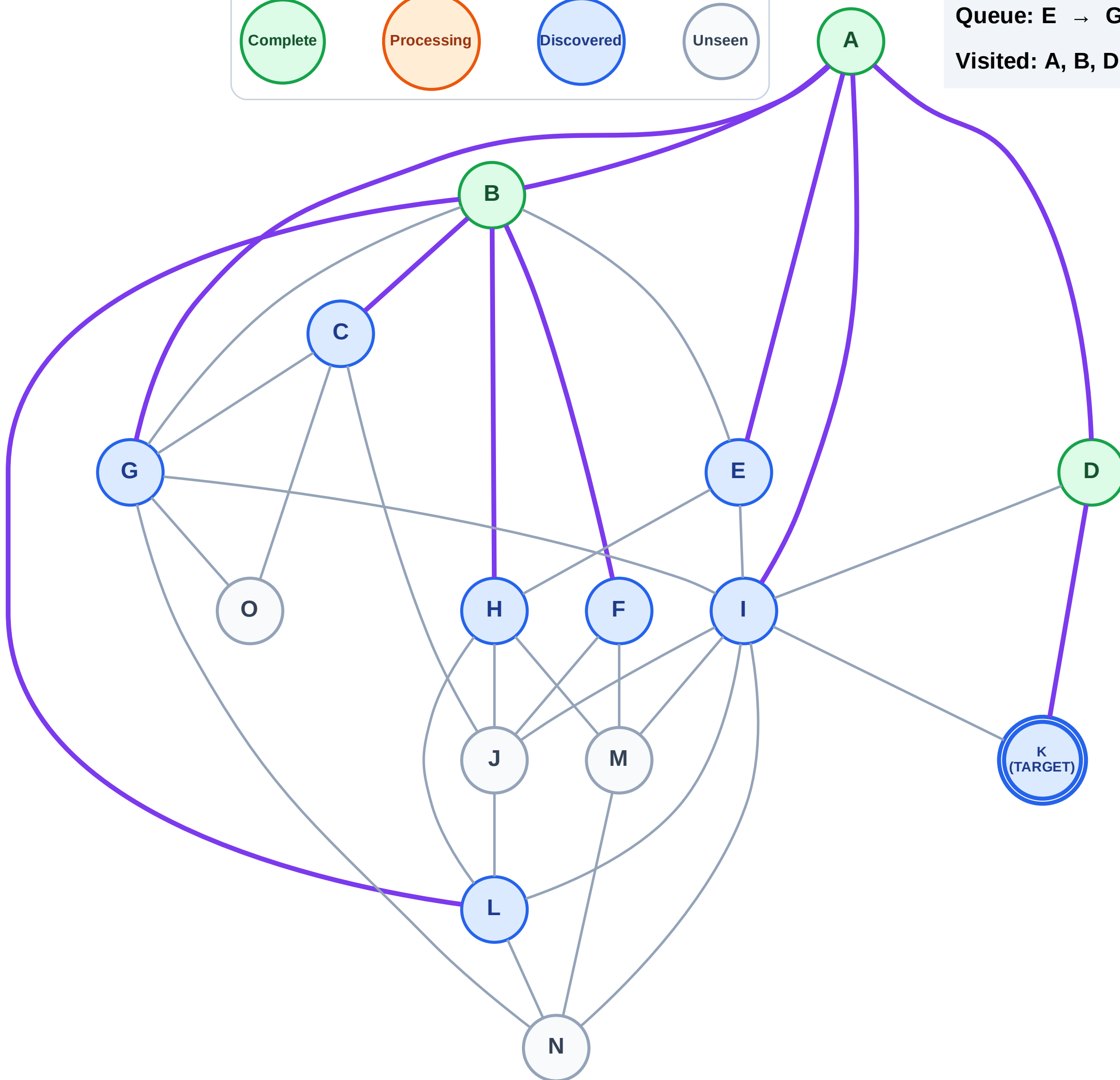
Step 7: Discover K from D

Legend



Queue: E → G → I → C → F → H → L → K

Visited: A, B, D



BFS Traversal

Step 8: Process node E

Legend

Complete

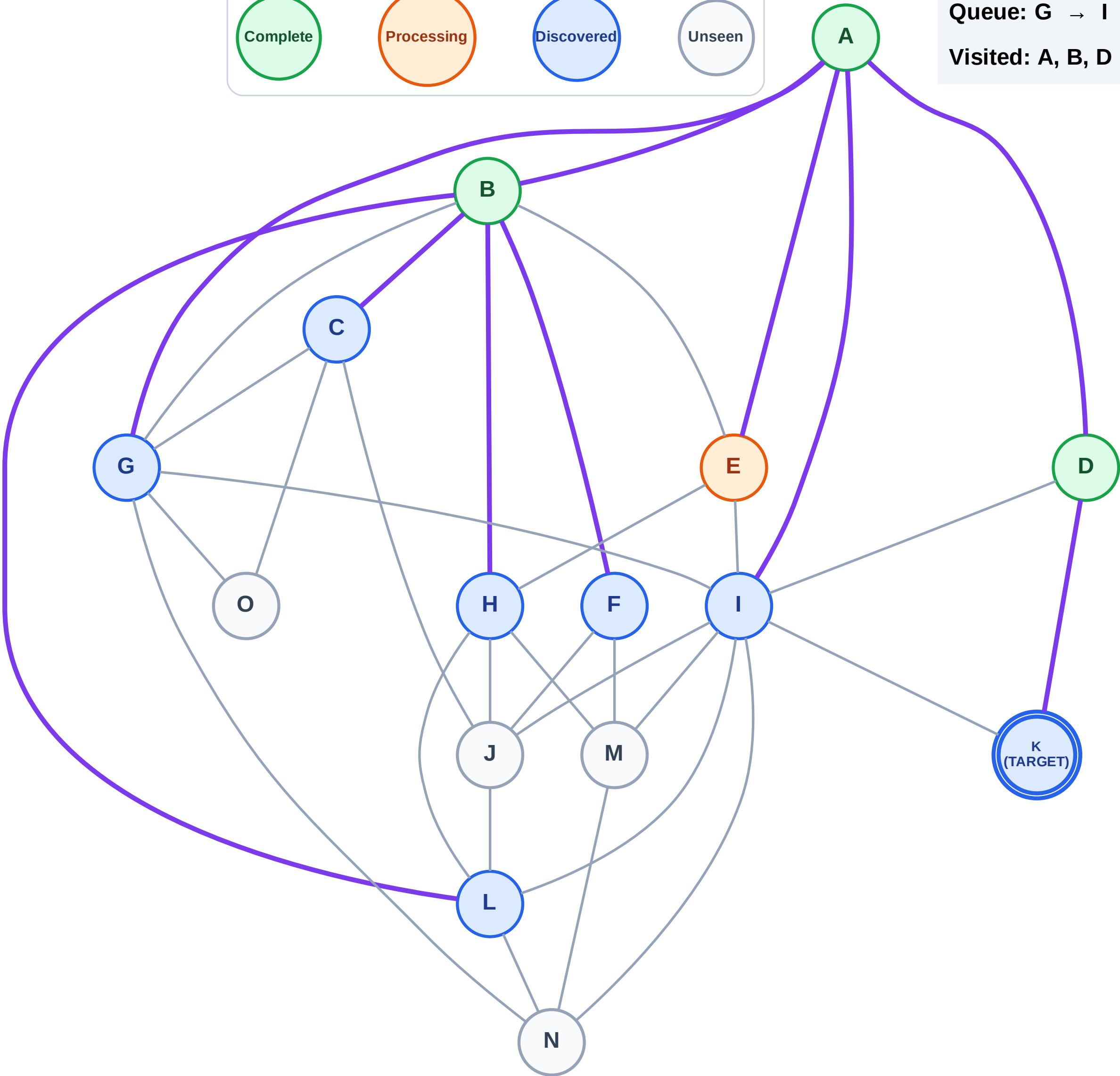
Processing

Discovered

Unseen

Queue: G → I → C → F → H → L → K

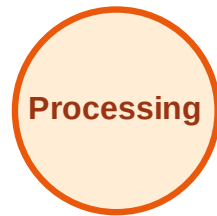
Visited: A, B, D



BFS Traversal

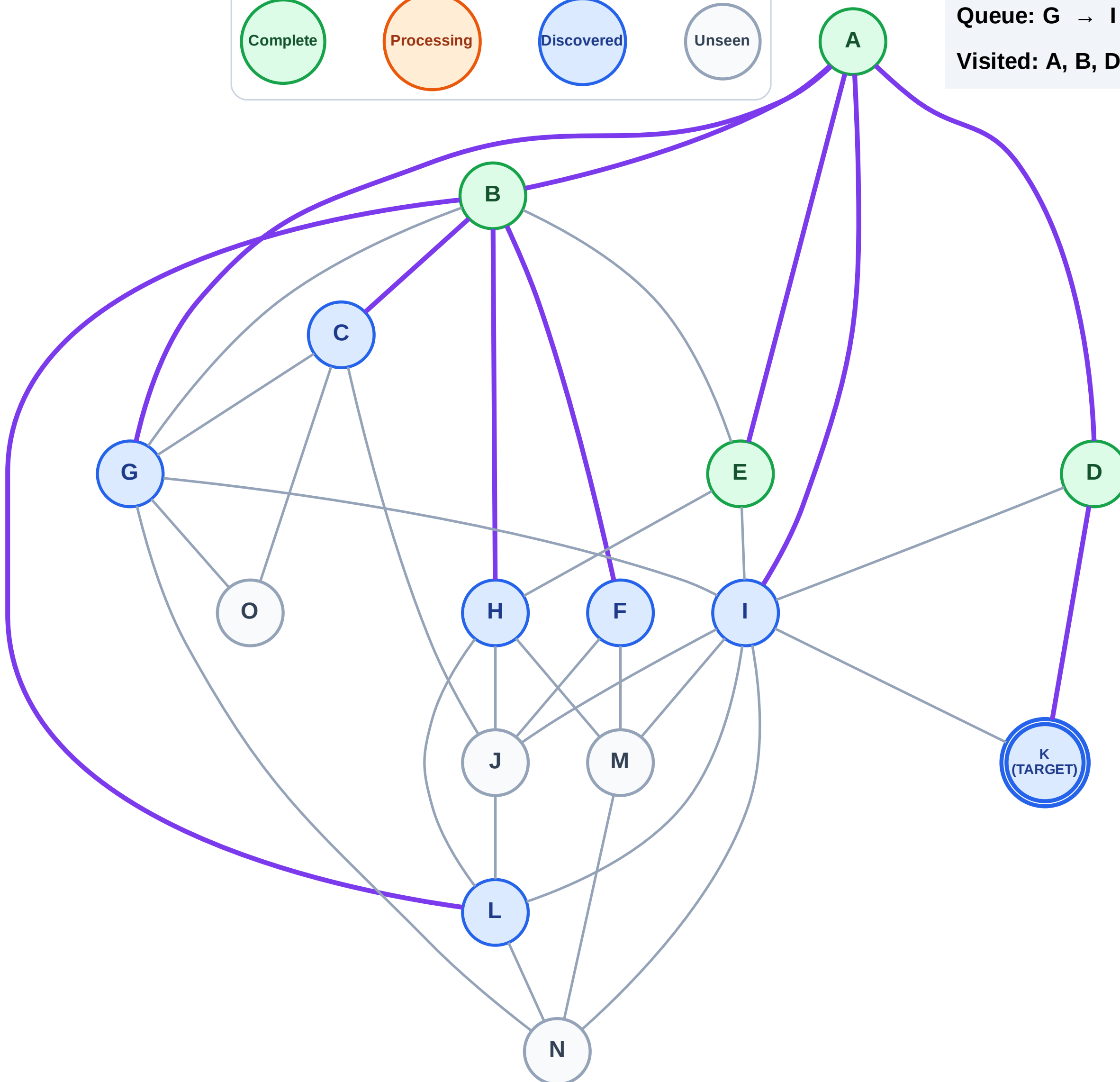
Step 9: No new neighbors from E

Legend



Queue: G → I → C → F → H → L → K

Visited: A, B, D, E



BFS Traversal

Step 10: Process node G

Legend



Complete



Processing



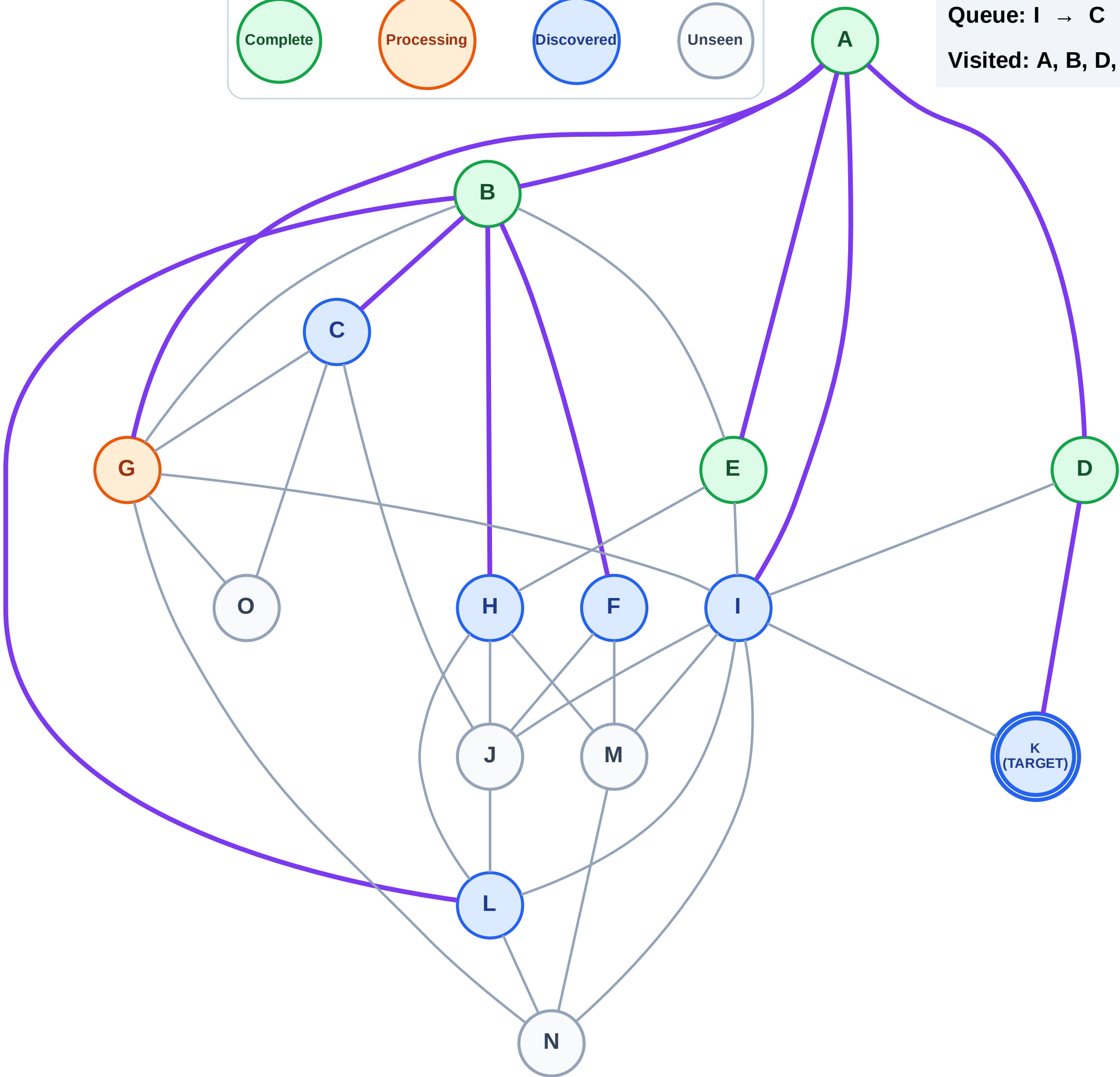
Discovered



Unseen

Queue: I → C → F → H → L → K

Visited: A, B, D, E



BFS Traversal

Step 11: Discover N, O from G

Legend



Complete



Processing



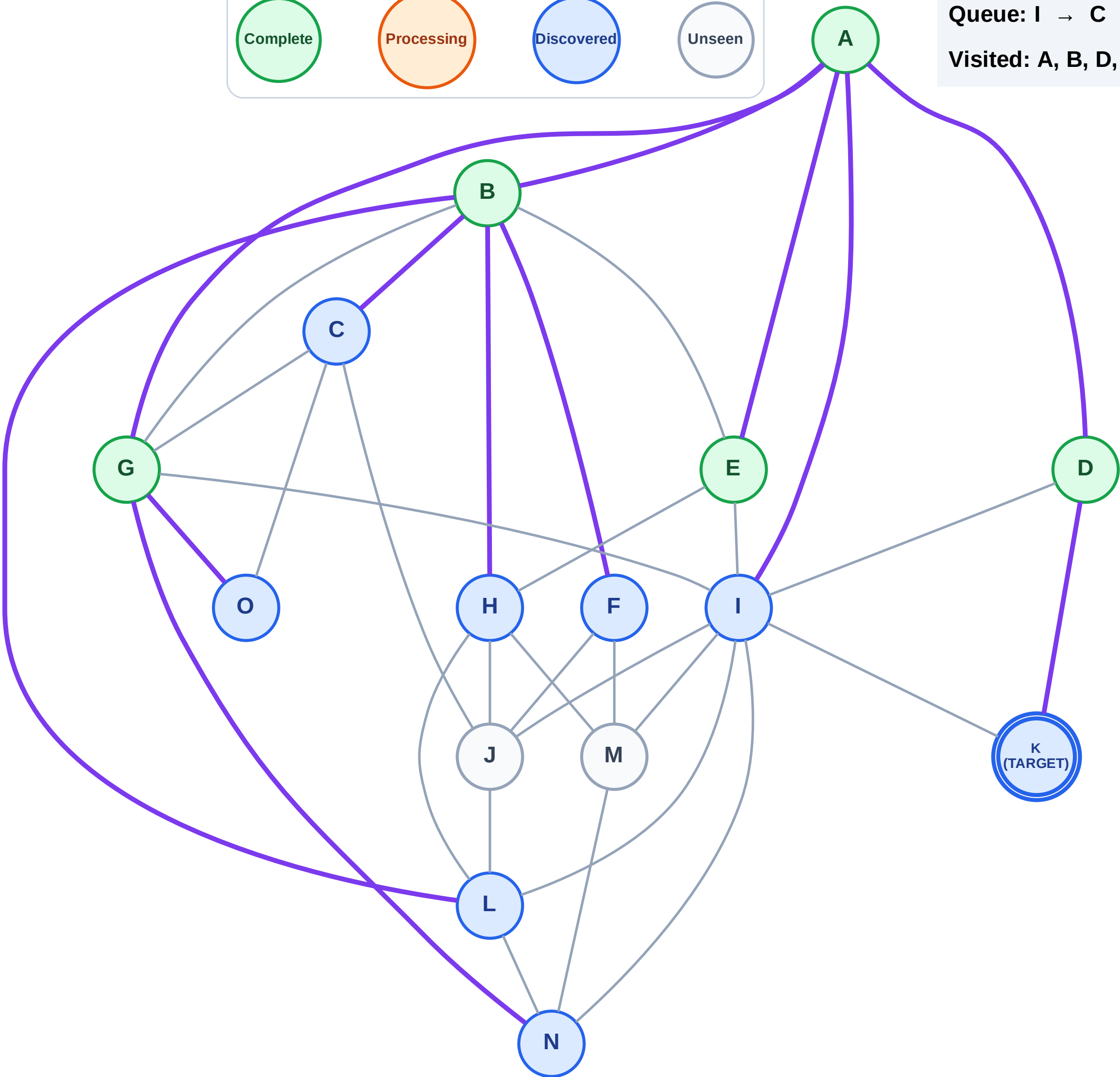
Discovered



Unseen

Queue: I → C → F → H → L → K → N → O

Visited: A, B, D, E, G



BFS Traversal

Step 12: Process node I

Legend

Complete

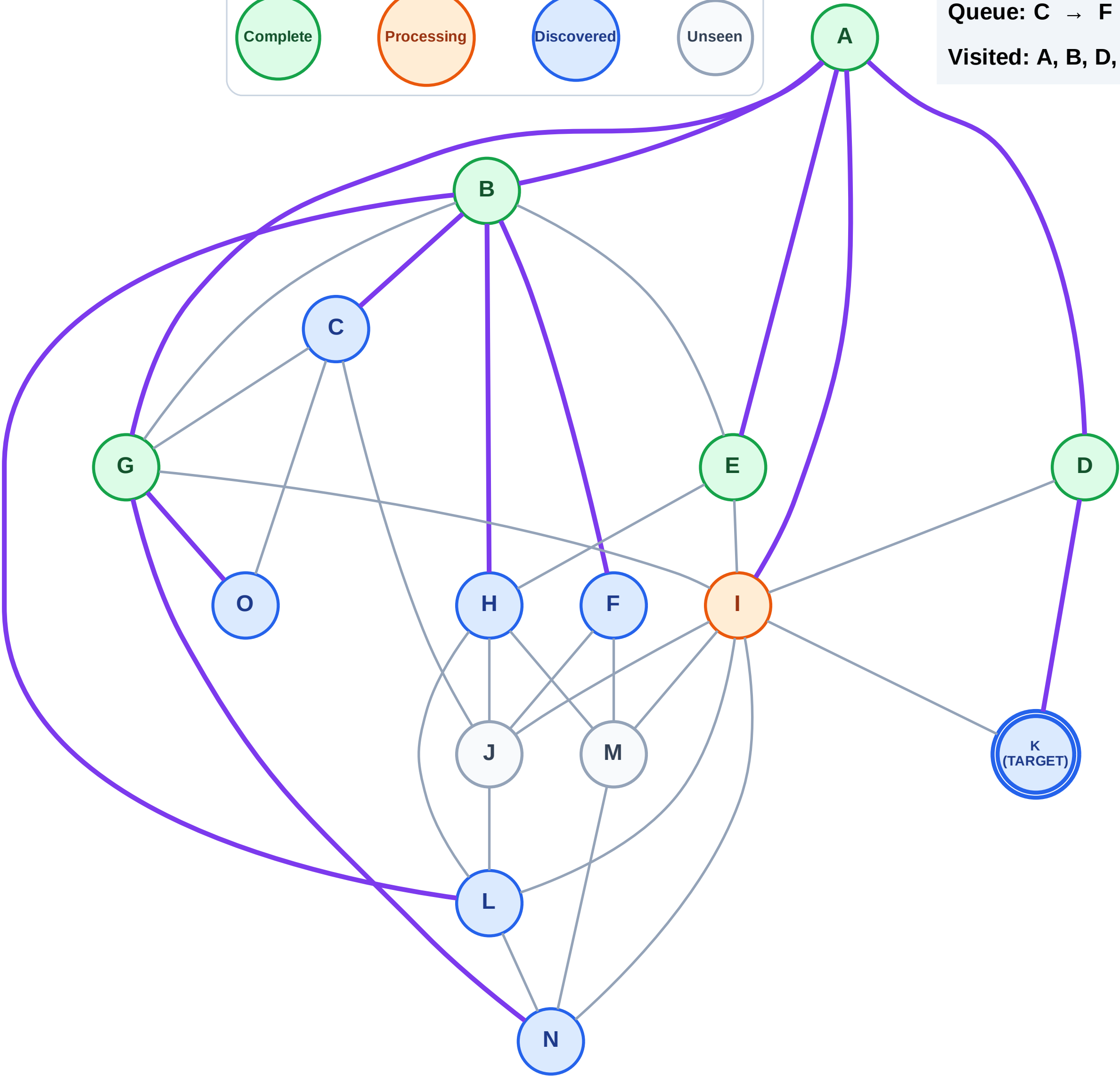
Processing

Discovered

Unseen

Queue: C → F → H → L → K → N → O

Visited: A, B, D, E, G



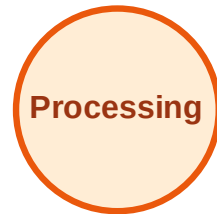
BFS Traversal

Step 13: Discover J, M from I

Legend



Complete



Processing



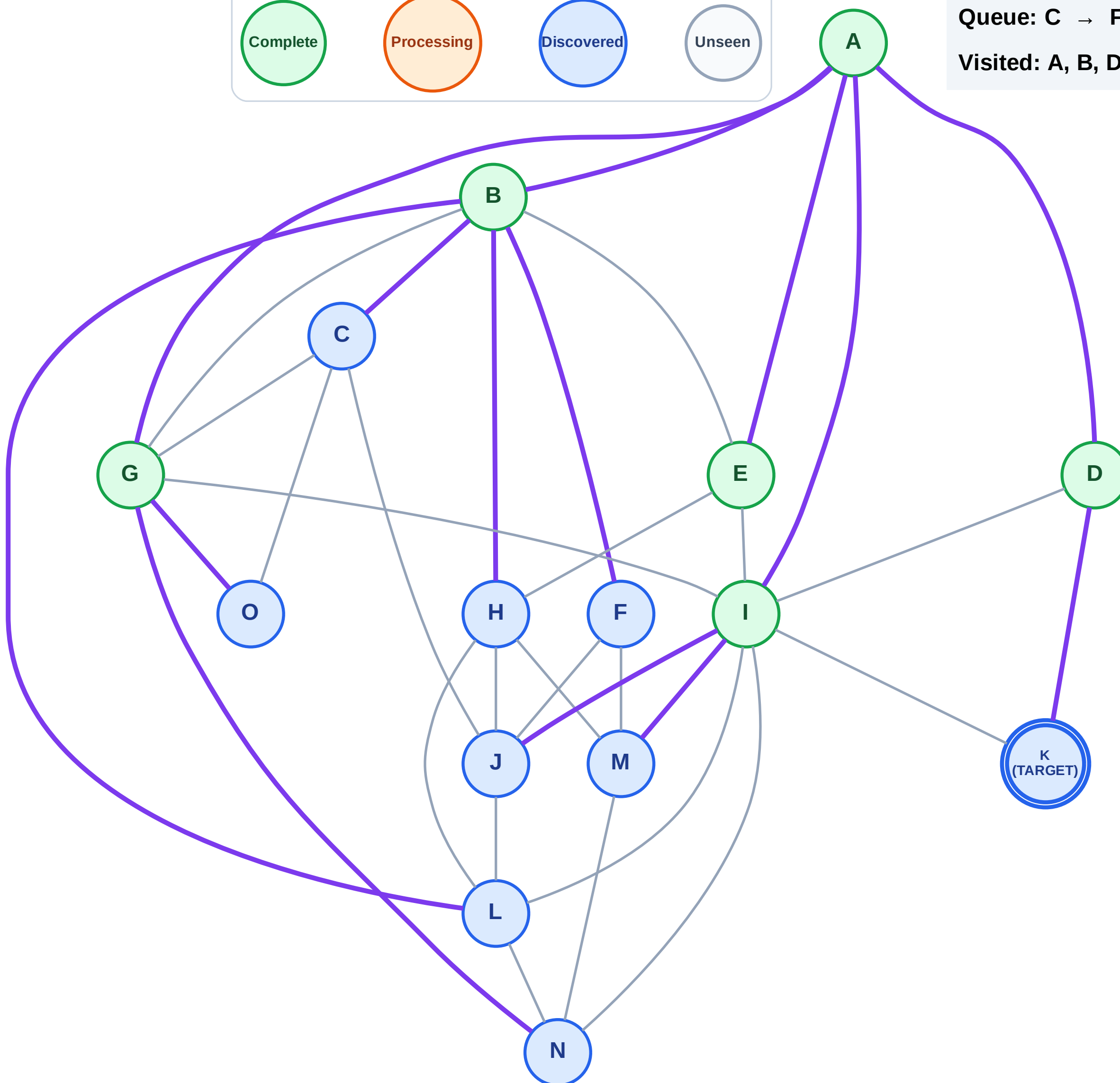
Discovered



Unseen

Queue: C → F → H → L → K → N → O → J → M

Visited: A, B, D, E, G, I



BFS Traversal

Step 14: Process node C

Legend

Complete

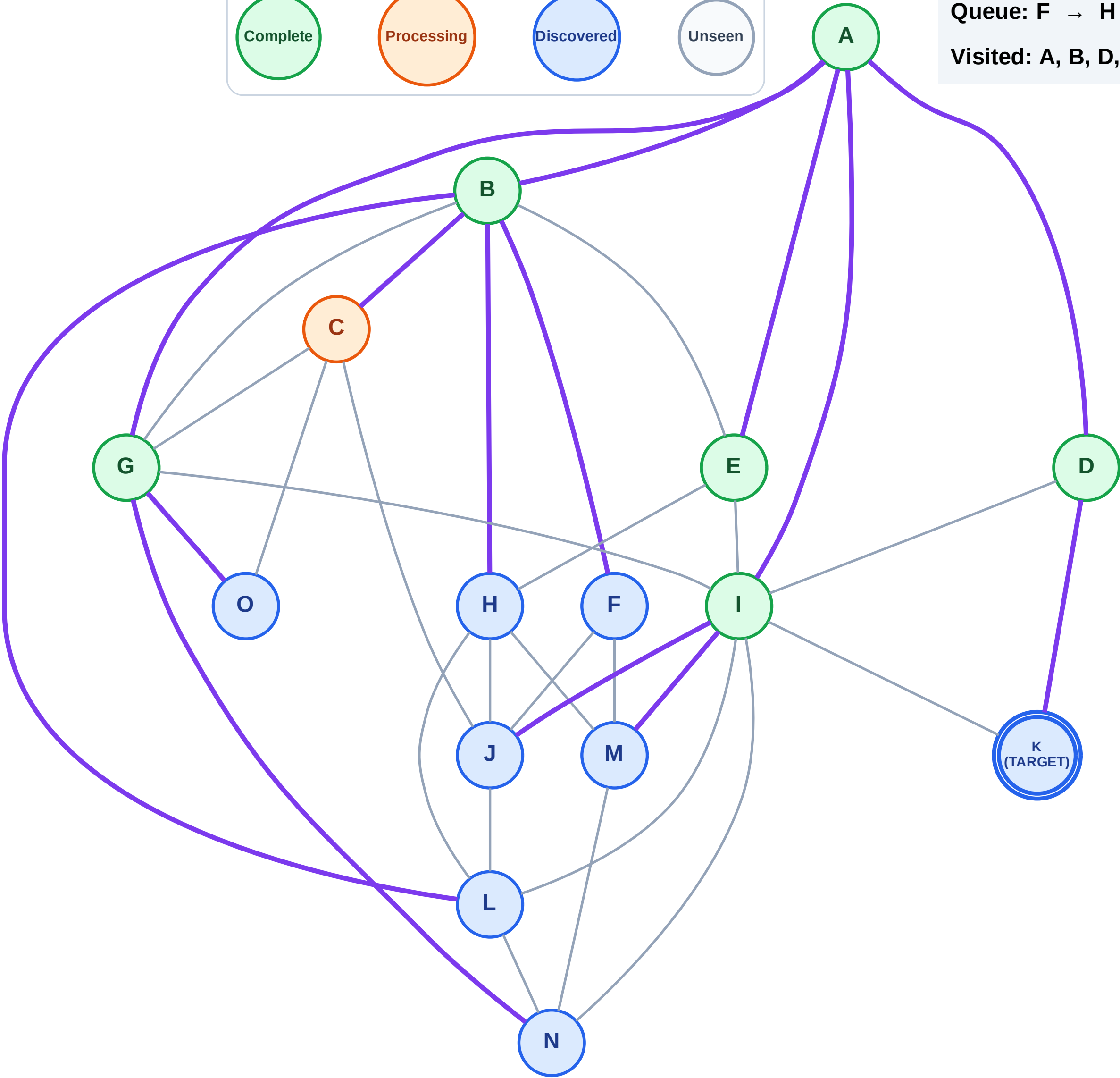
Processing

Discovered

Unseen

Queue: F → H → L → K → N → O → J → M

Visited: A, B, D, E, G, I



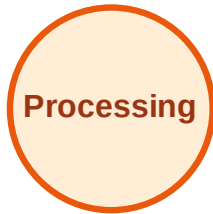
BFS Traversal

Step 15: No new neighbors from C

Legend



Complete



Processing



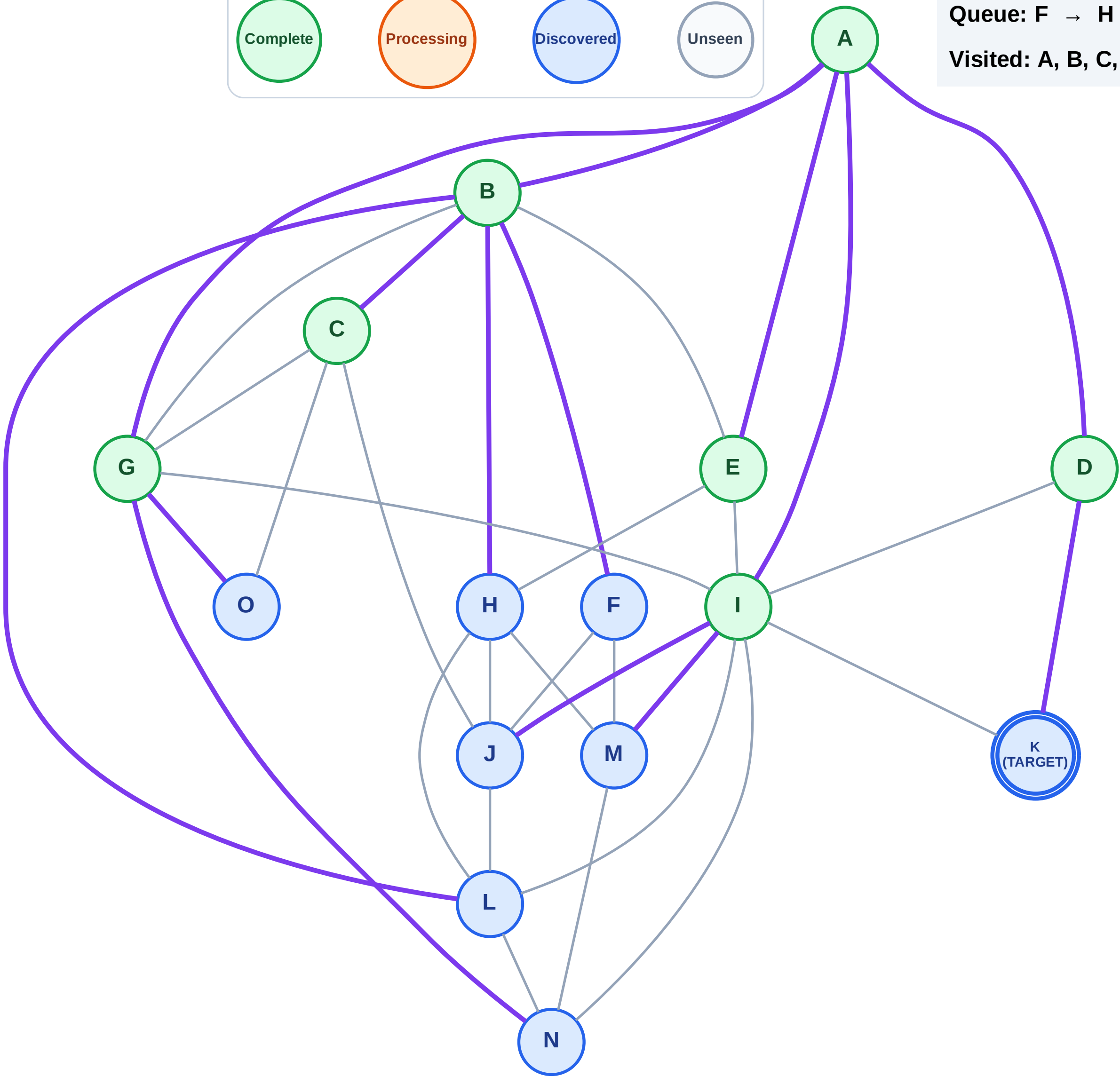
Discovered



Unseen

Queue: F → H → L → K → N → O → J → M

Visited: A, B, C, D, E, G, I



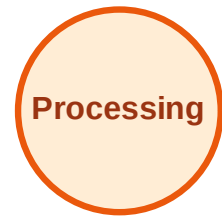
BFS Traversal

Step 16: Process node F

Legend



Complete



Processing



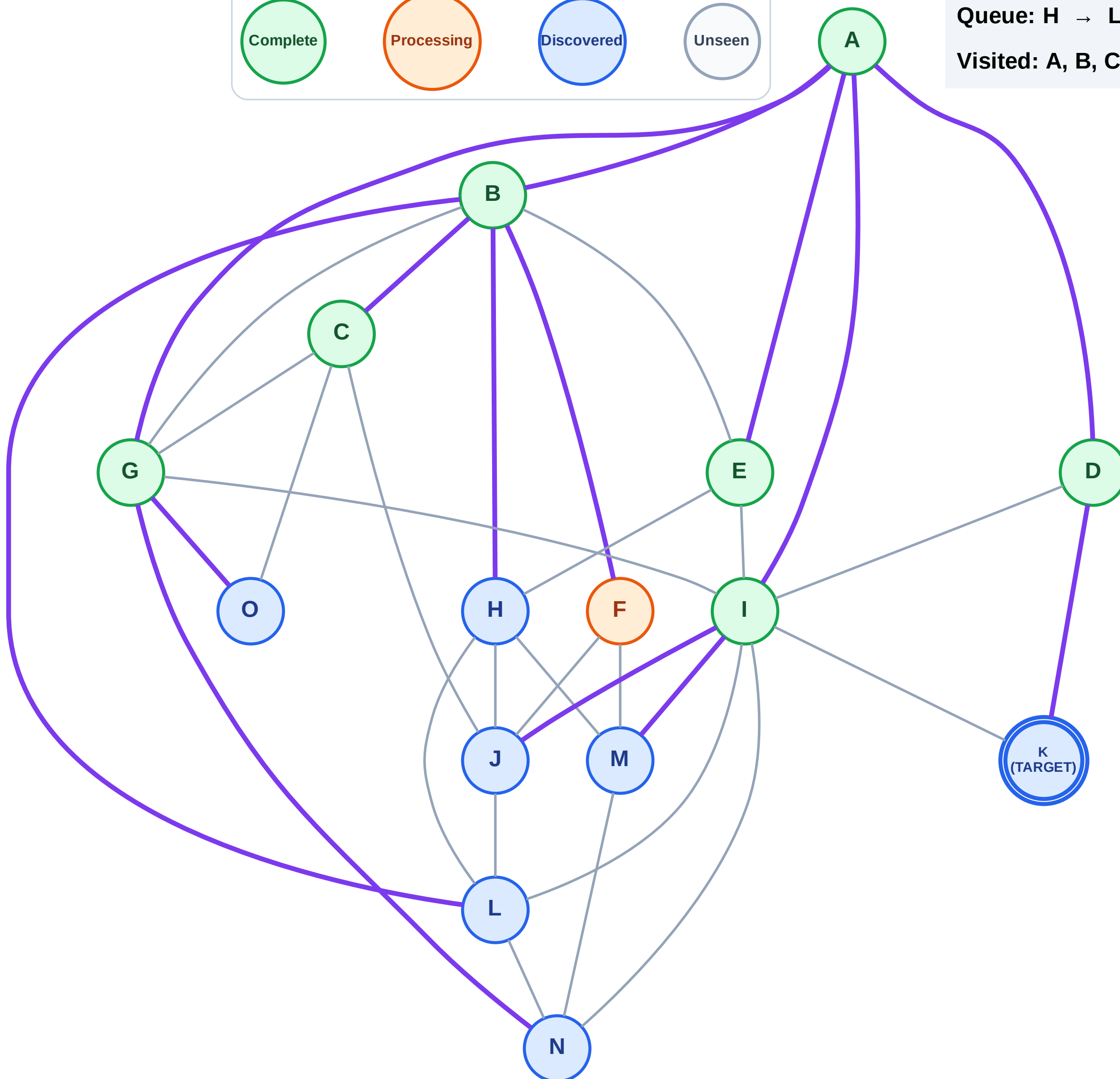
Discovered



Unseen

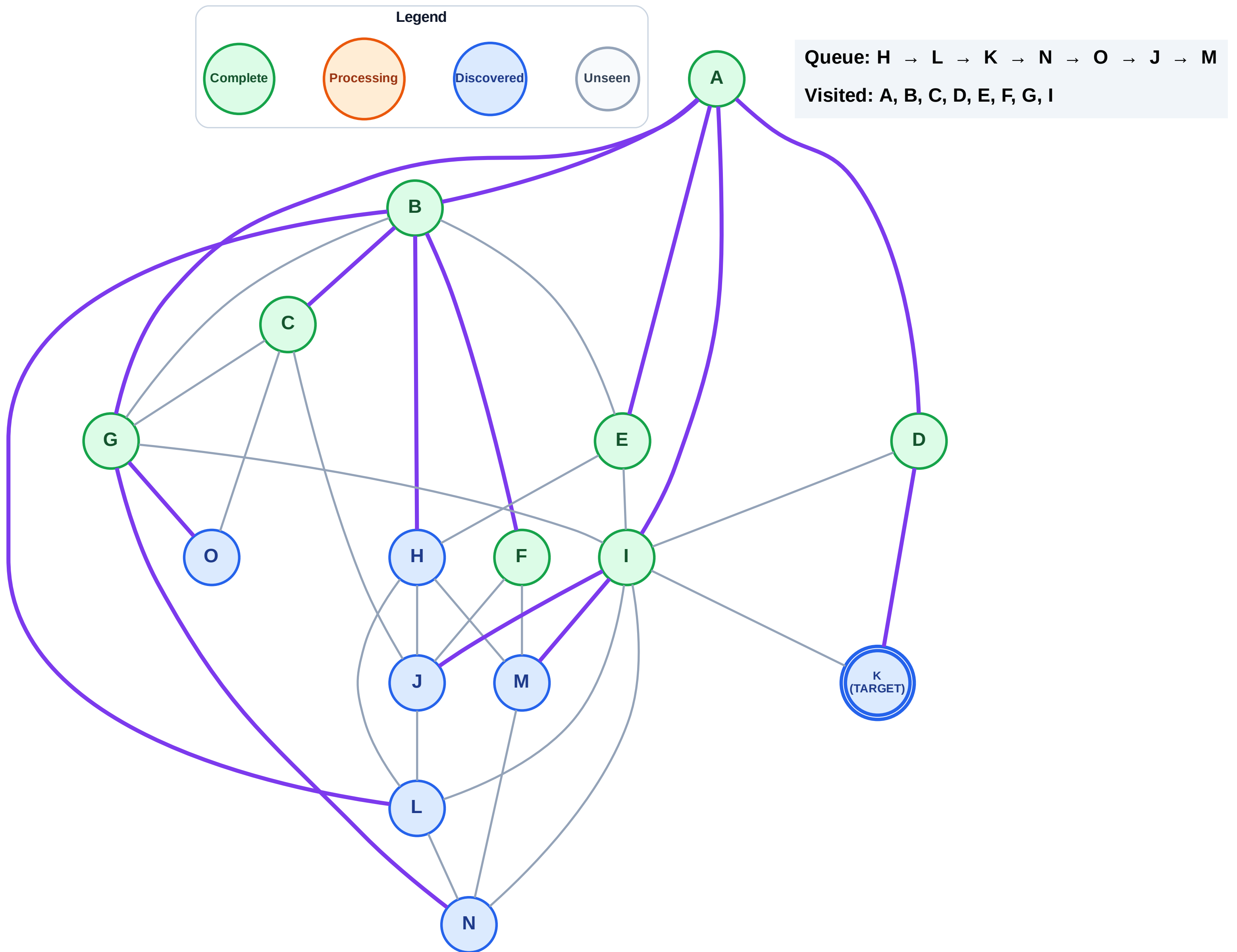
Queue: H → L → K → N → O → J → M

Visited: A, B, C, D, E, G, I



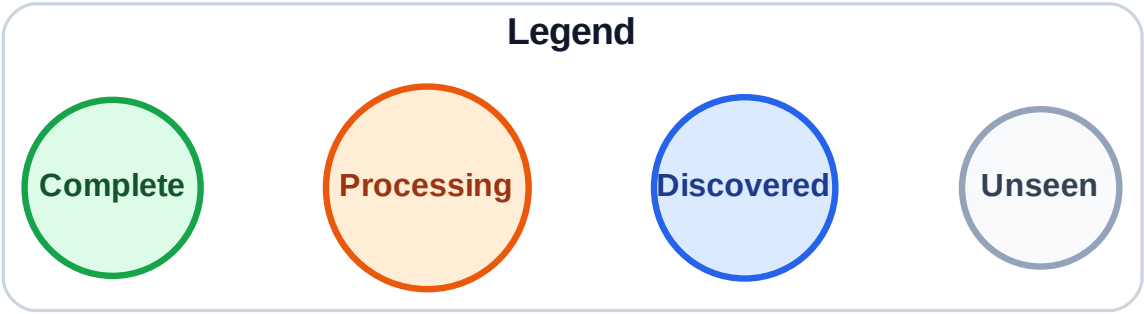
Step 17: No new neighbors from F

Step 17: No new neighbors from F



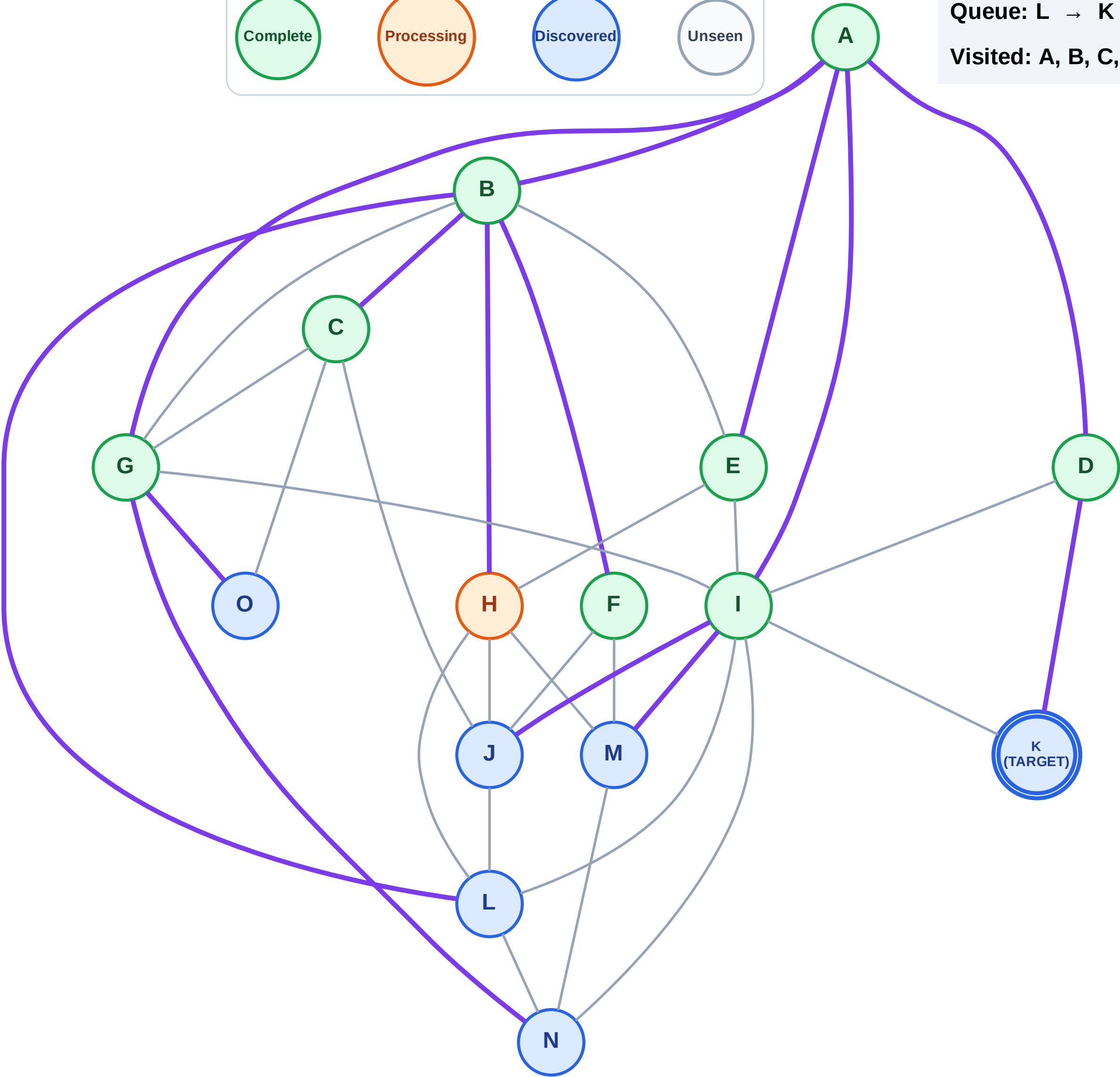
BFS Traversal

Step 18: Process node H



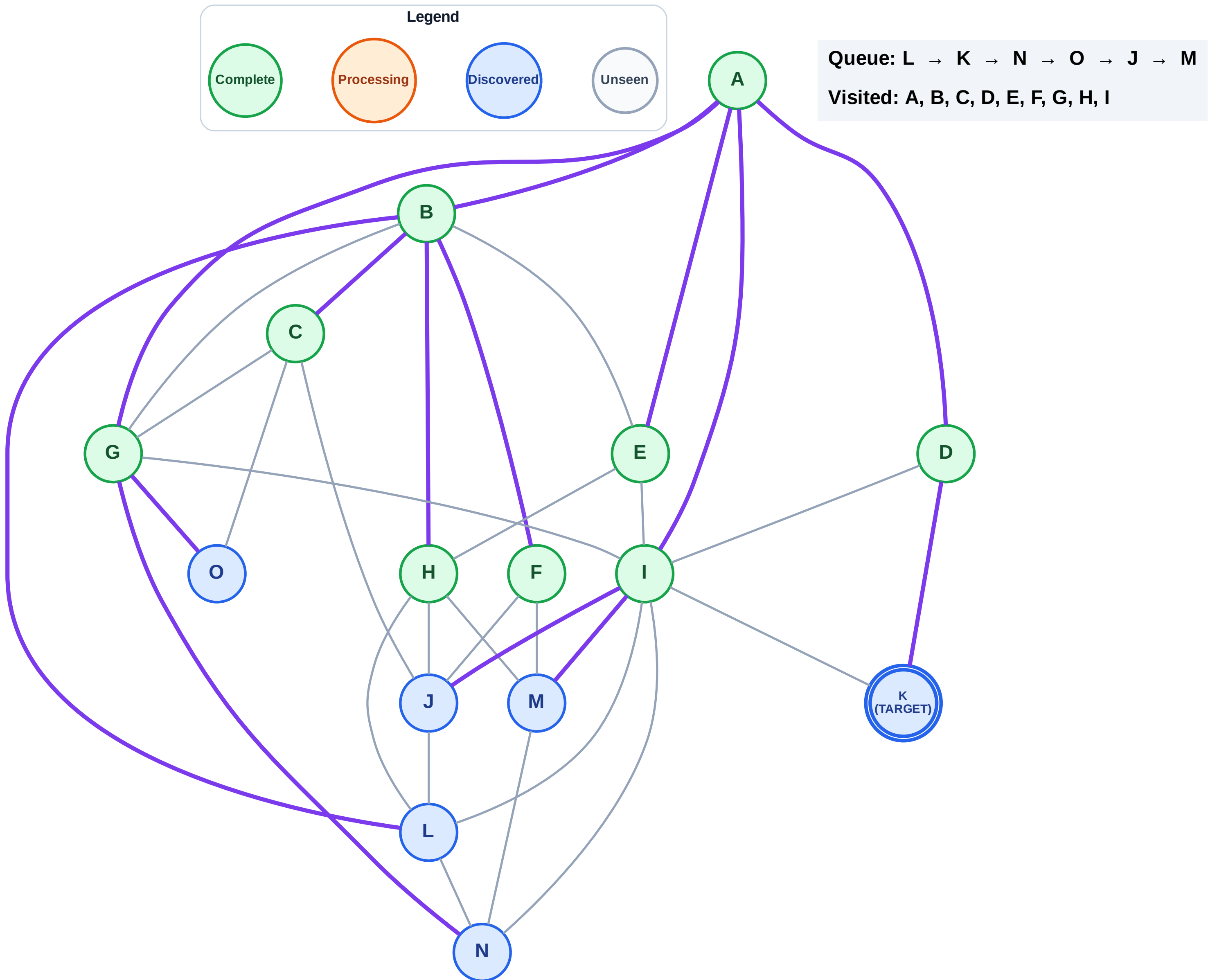
Queue: L → K → N → O → J → M

Visited: A, B, C, D, E, F, G, I



Step 19: No new neighbors from H

Step 19: No new neighbors from H



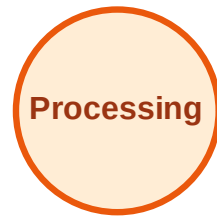
BFS Traversal

Step 20: Process node L

Legend



Complete



Processing



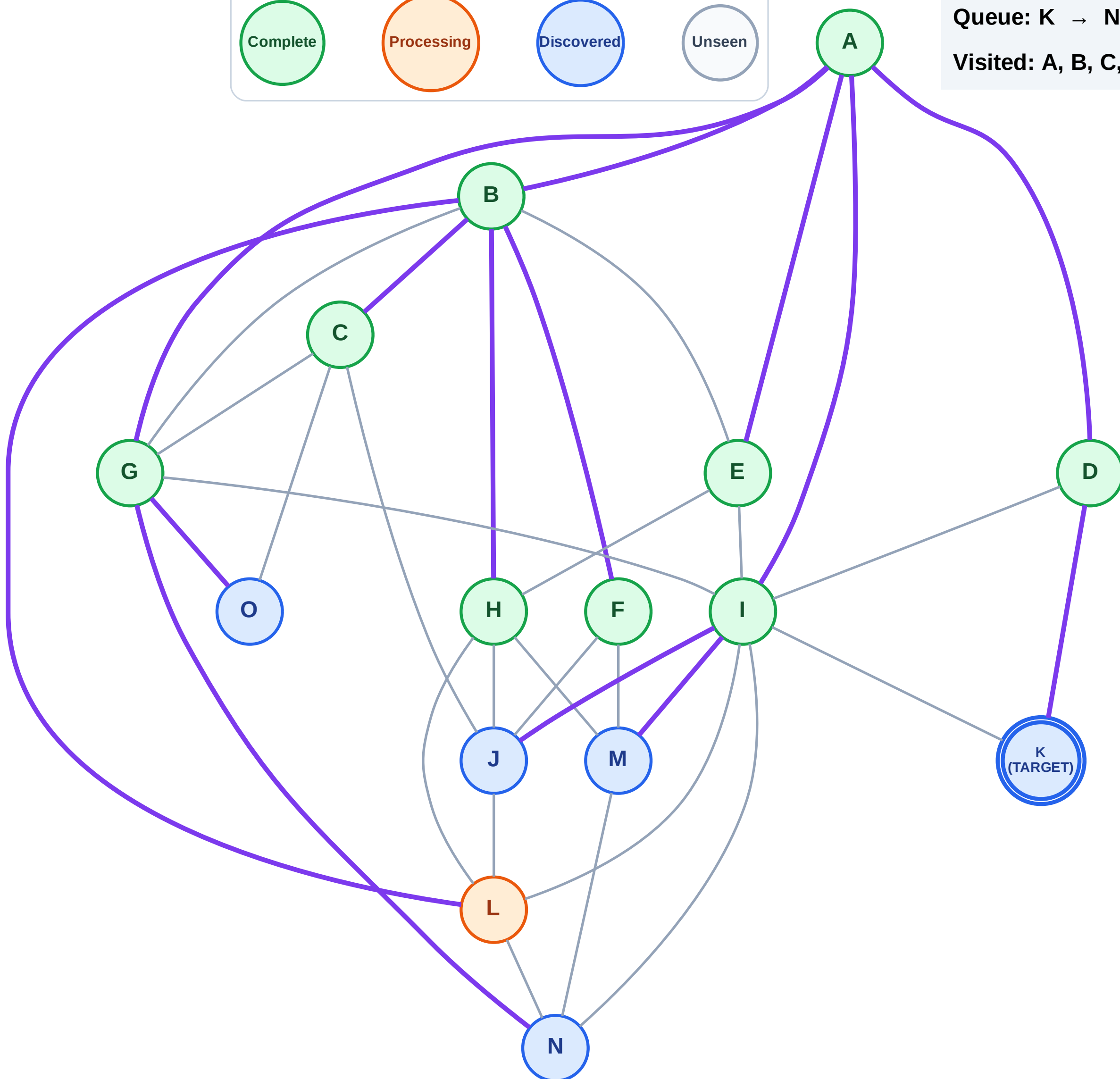
Discovered



Unseen

Queue: K → N → O → J → M

Visited: A, B, C, D, E, F, G, H, I



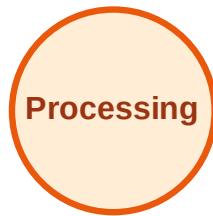
BFS Traversal

Step 21: No new neighbors from L

Legend



Complete



Processing



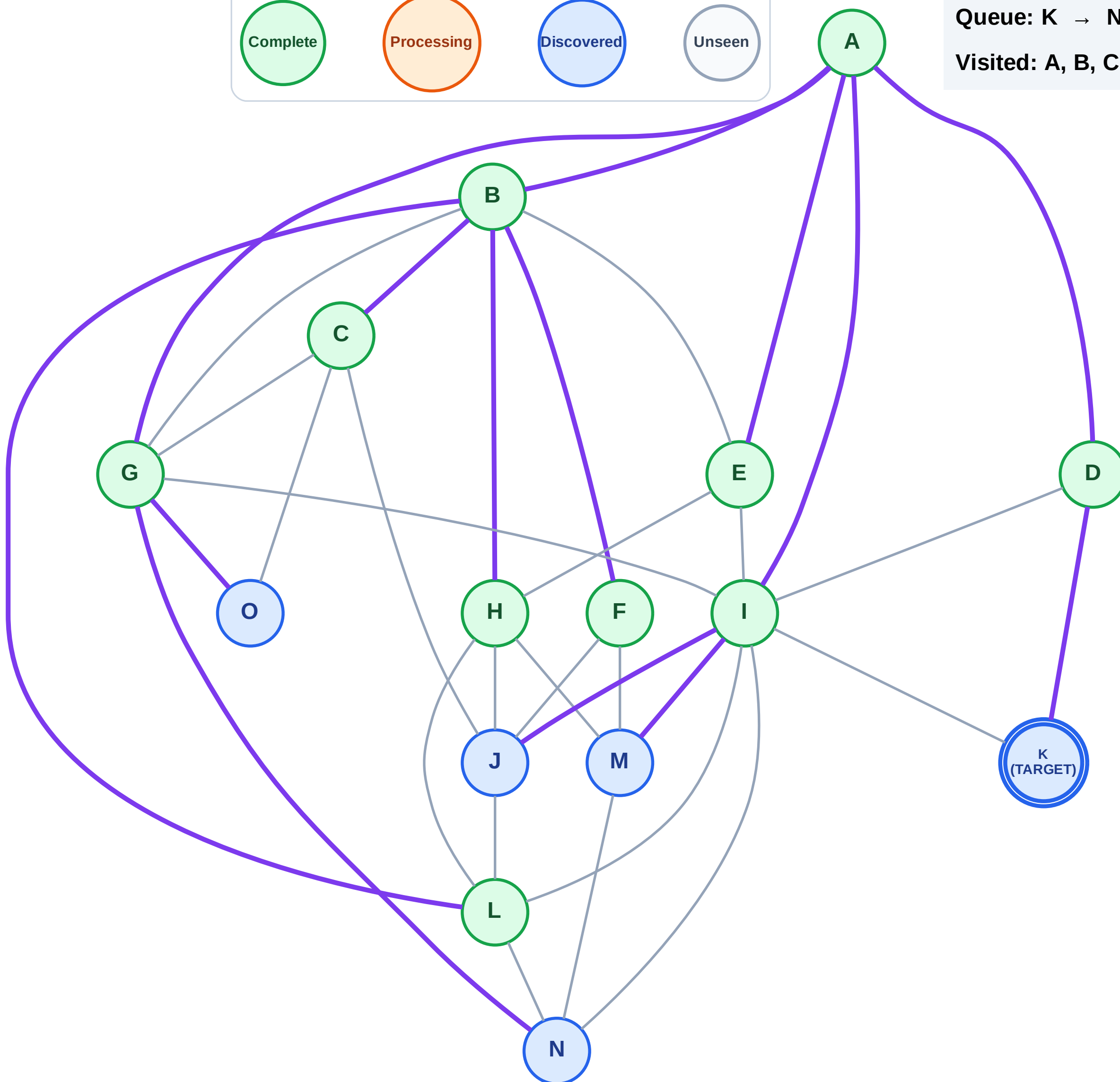
Discovered



Unseen

Queue: K → N → O → J → M

Visited: A, B, C, D, E, F, G, H, I, L



BFS Traversal

Step 22: Process node K

Legend



Complete



Processing



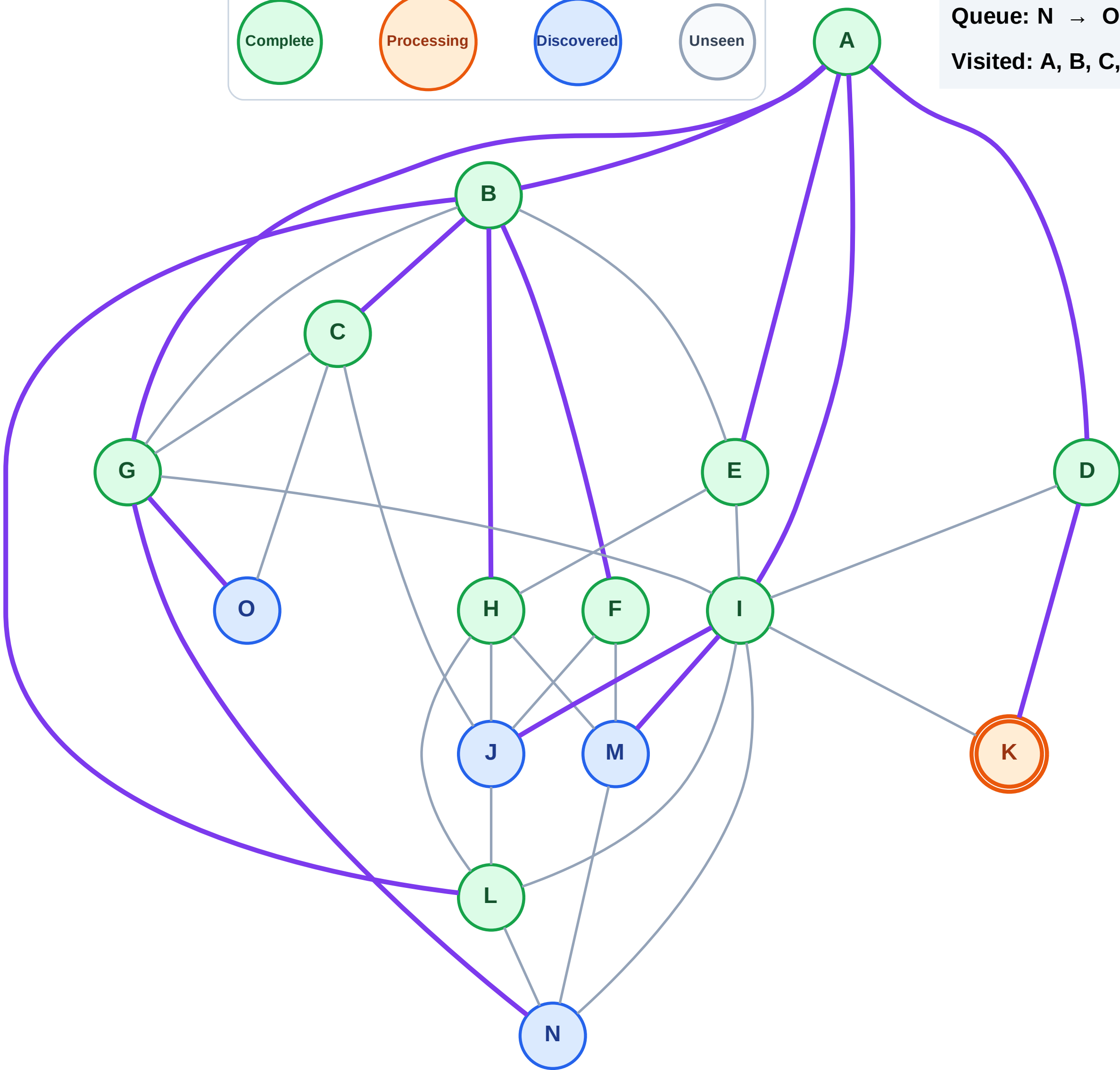
Discovered



Unseen

Queue: N → O → J → M

Visited: A, B, C, D, E, F, G, H, I, L



BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend

Complete

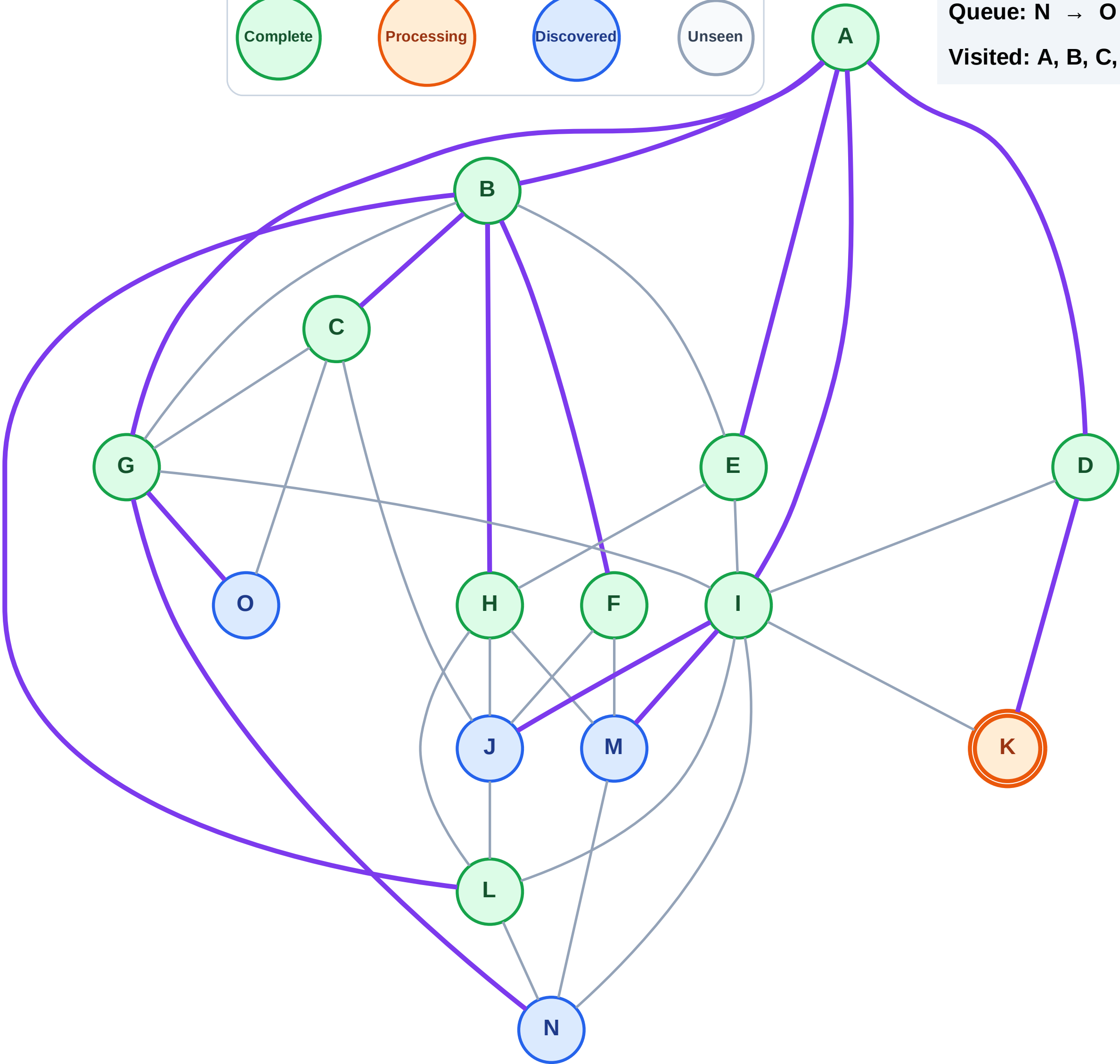
Processing

Discovered

Unseen

Queue: N → O → J → M

Visited: A, B, C, D, E, F, G, H, I, K, L



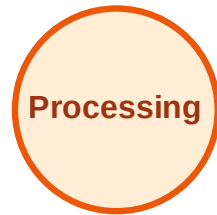
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



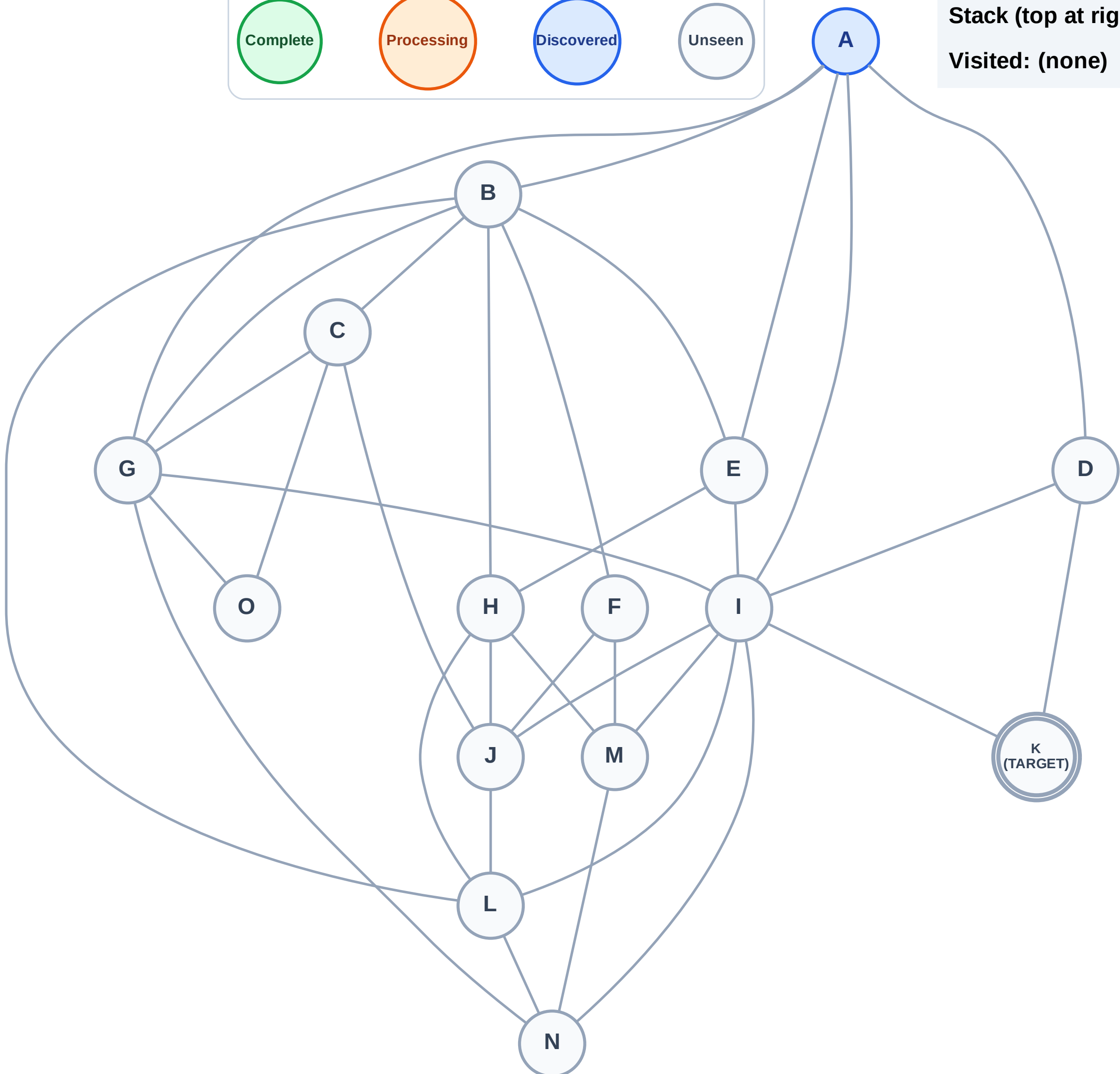
Discovered



Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

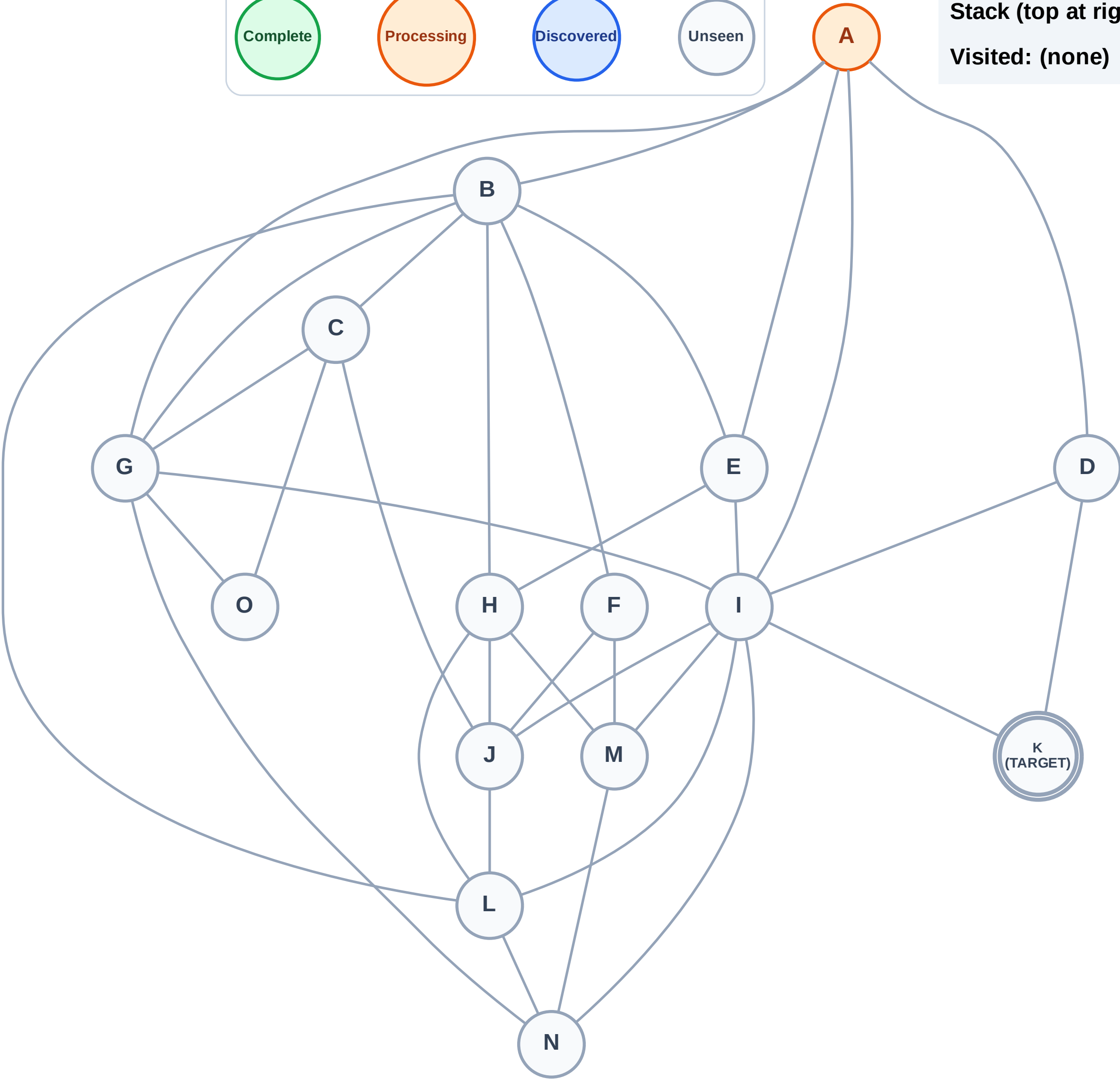
Complete

Processing

Discovered

Unseen

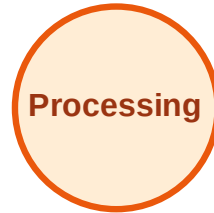
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

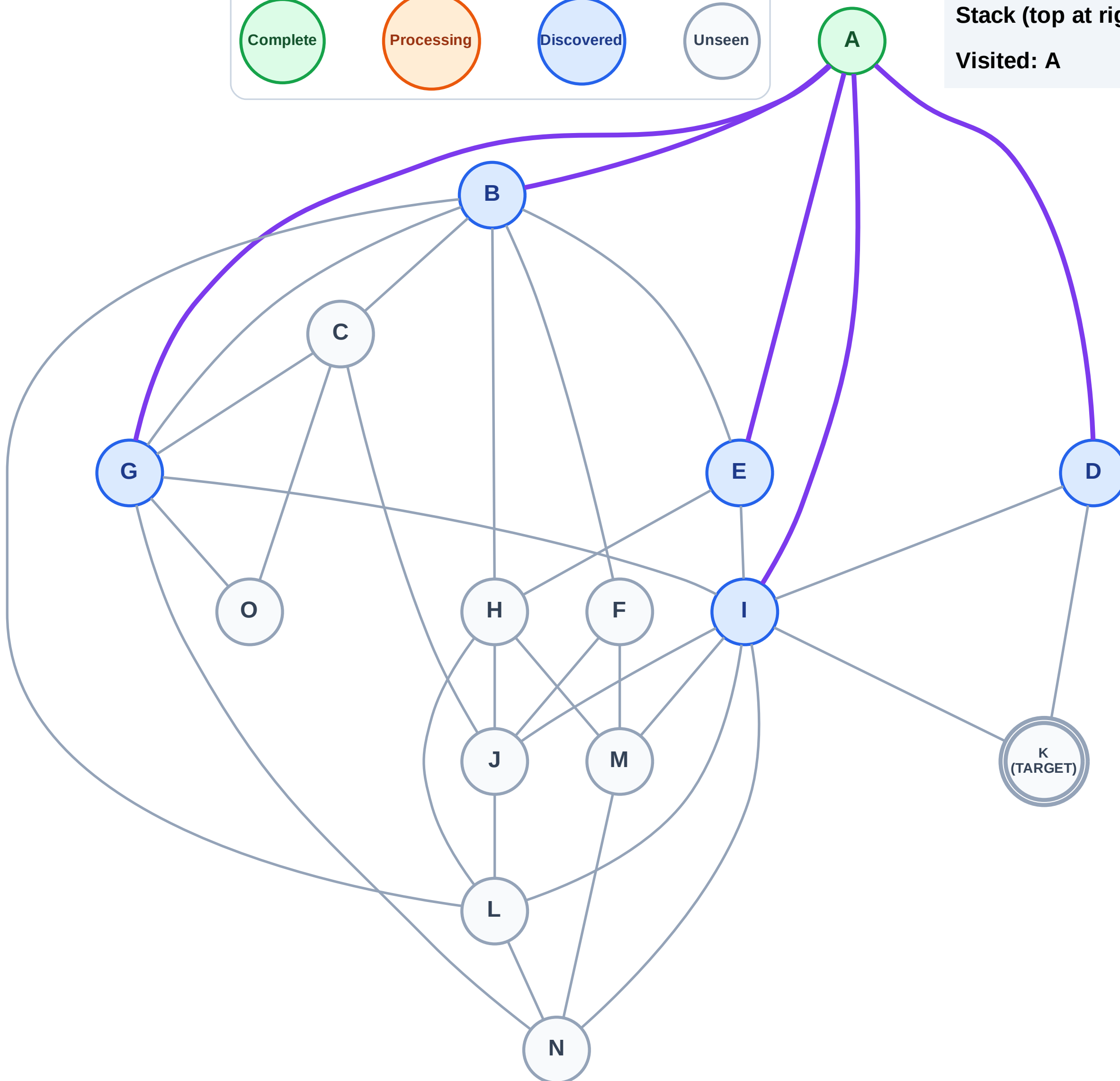
Step 3: Discover I, G, E, D, B from A

Legend



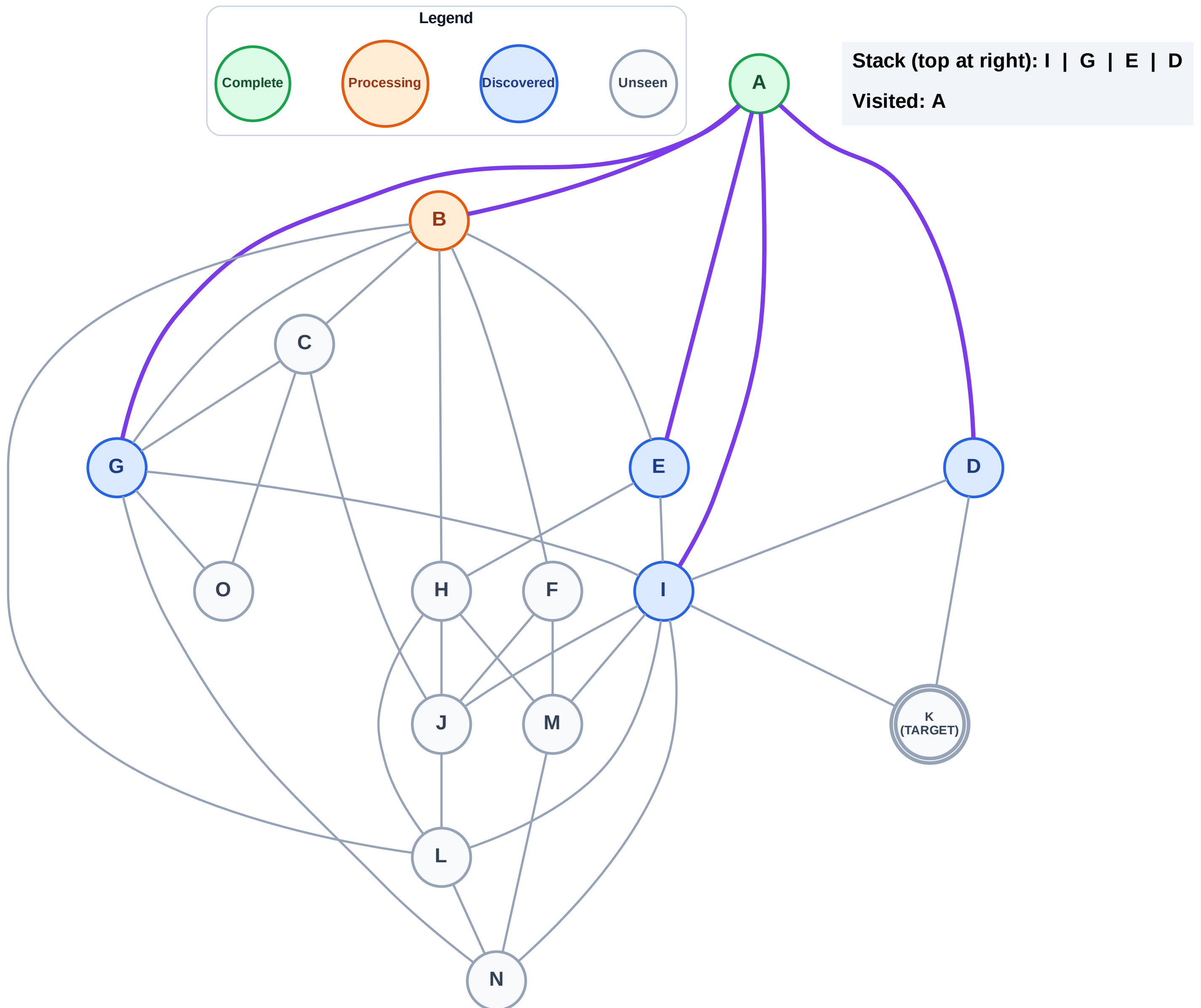
Stack (top at right): I | G | E | D | B

Visited: A



Step 4: Process node B

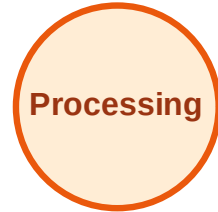
Step 4: Process node B



DFS Traversal

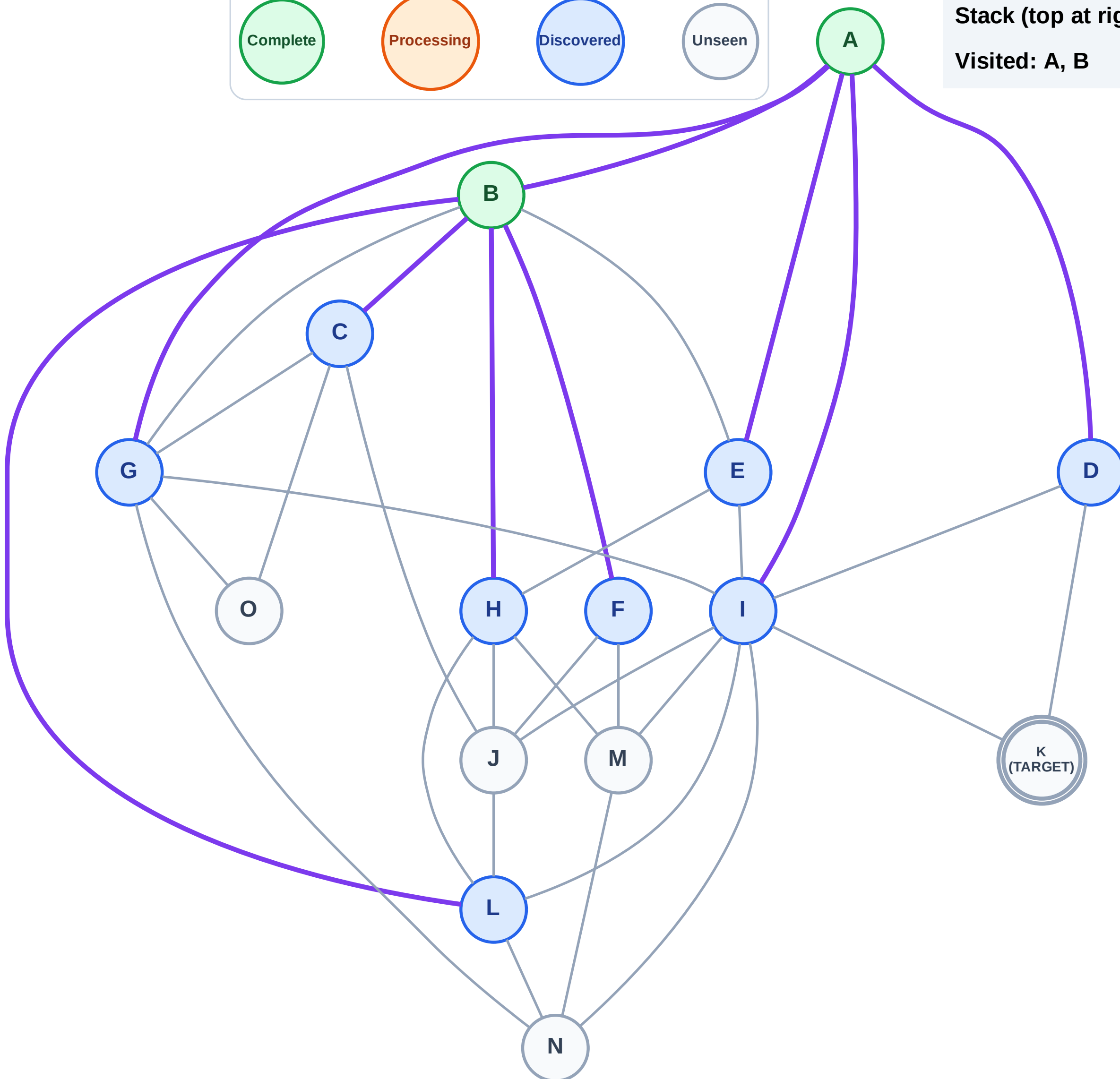
Step 5: Discover L, H, F, C from B

Legend



Stack (top at right): I | G | E | D | L | H | F | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

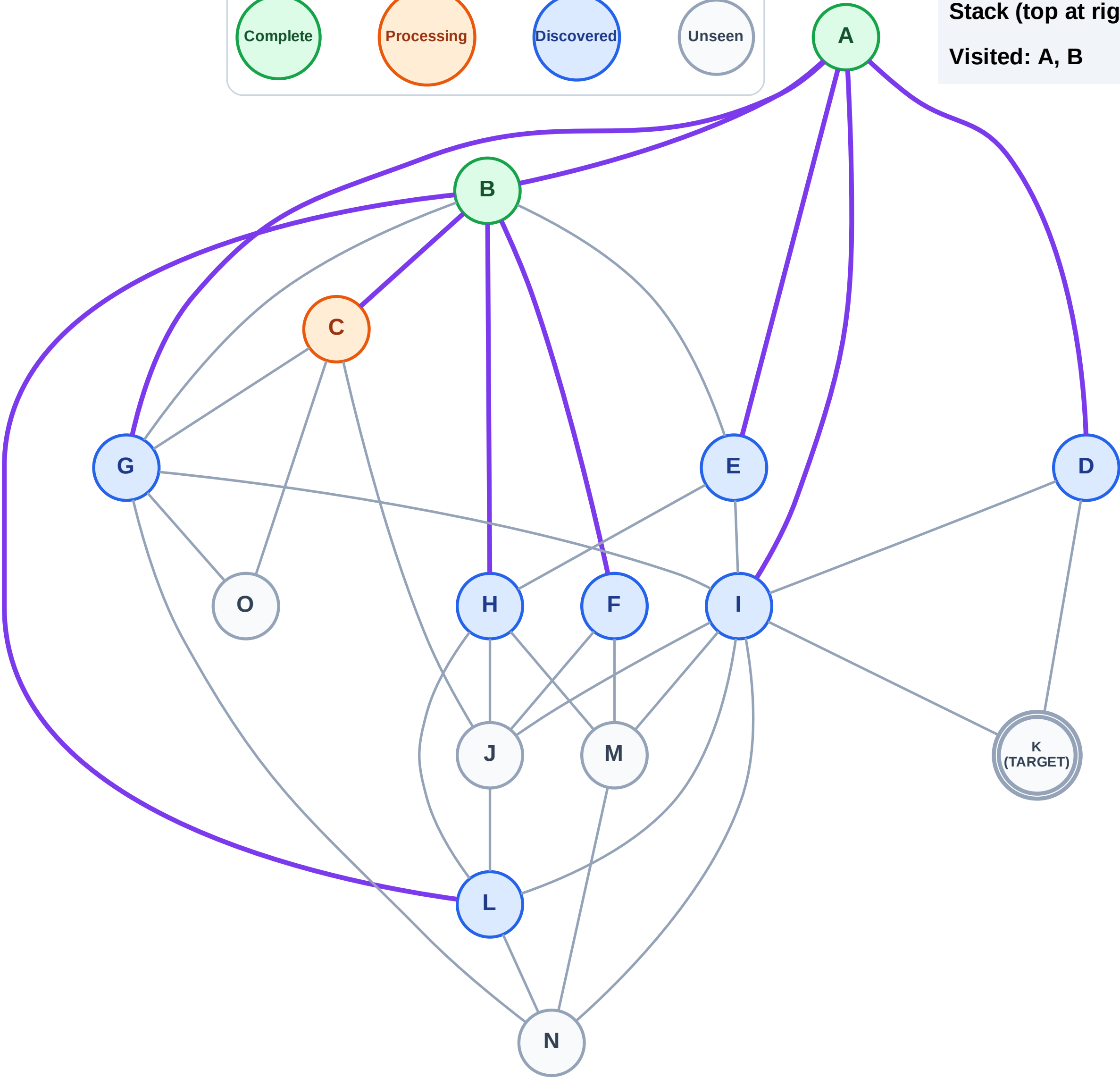
Processing

Discovered

Unseen

Stack (top at right): I | G | E | D | L | H | F

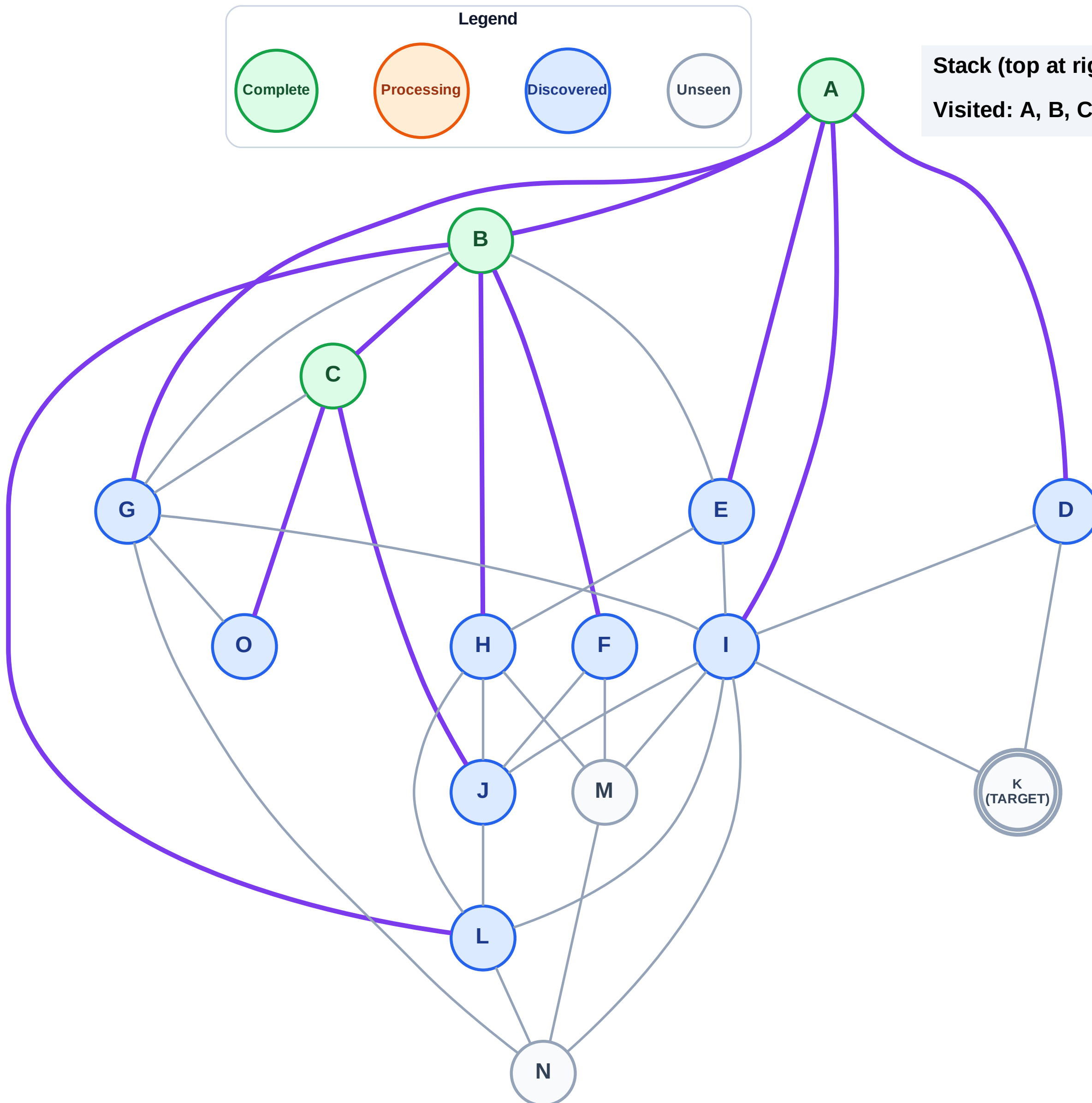
Visited: A, B



Step 7: Discover O, J from C

Stack (top at right): I | G | E | D | L | H | F | O | J

Visited: A, B, C



DFS Traversal

Step 8: Process node J

Legend

Complete

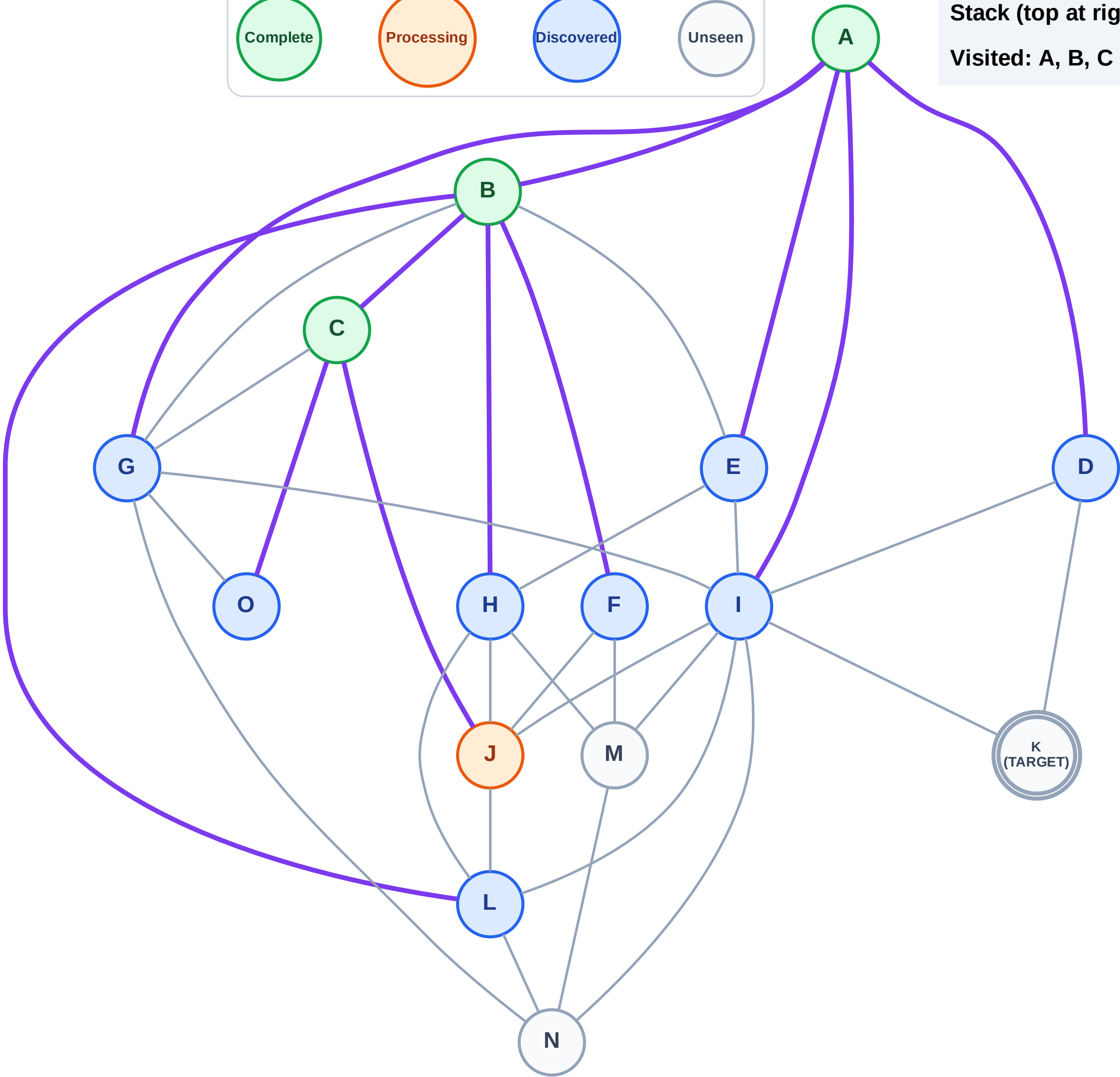
Processing

Discovered

Unseen

Stack (top at right): I | G | E | D | L | H | F | O

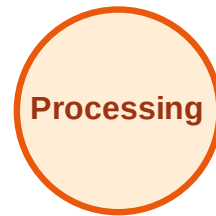
Visited: A, B, C



DFS Traversal

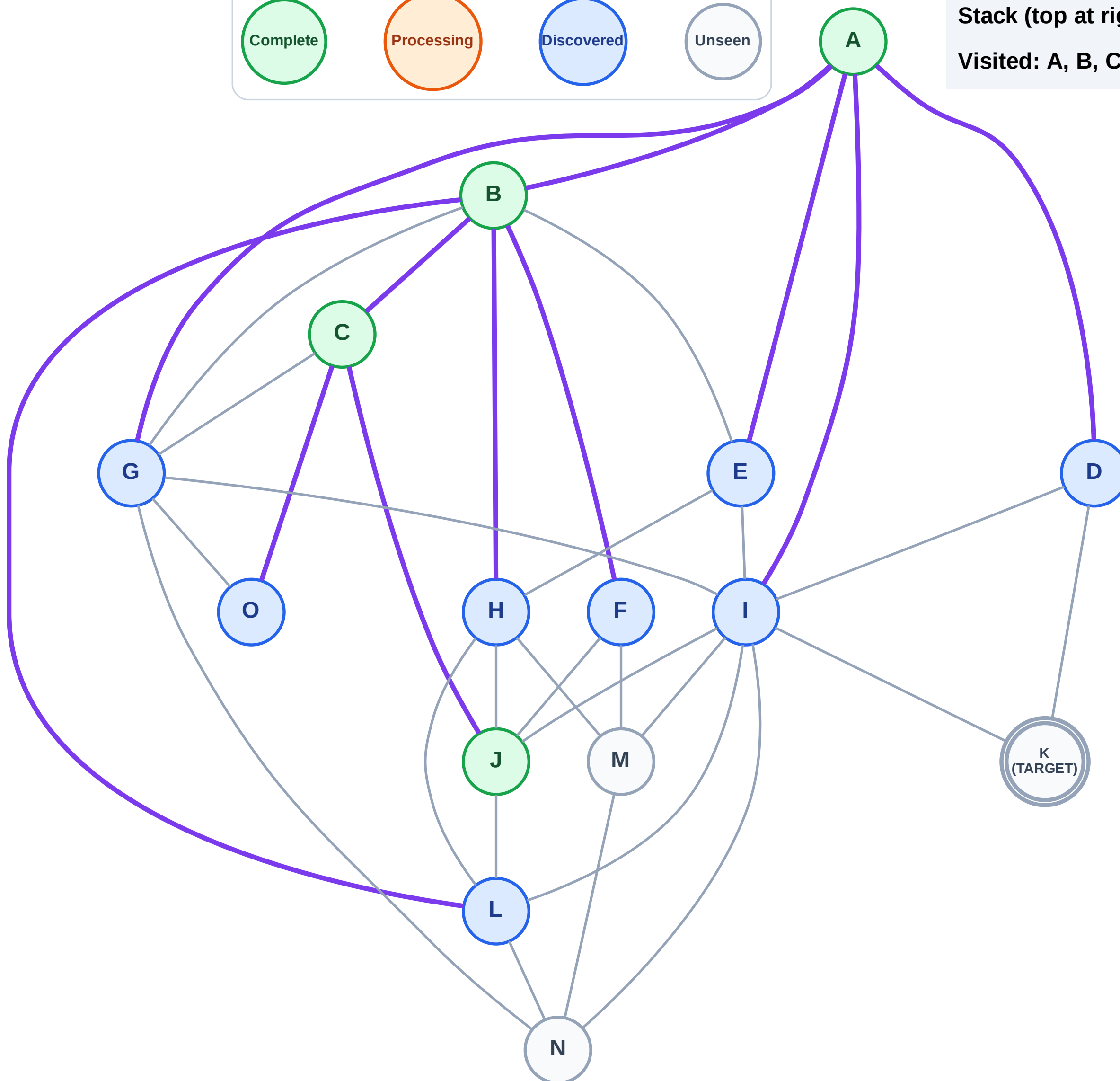
Step 9: No new neighbors from J

Legend



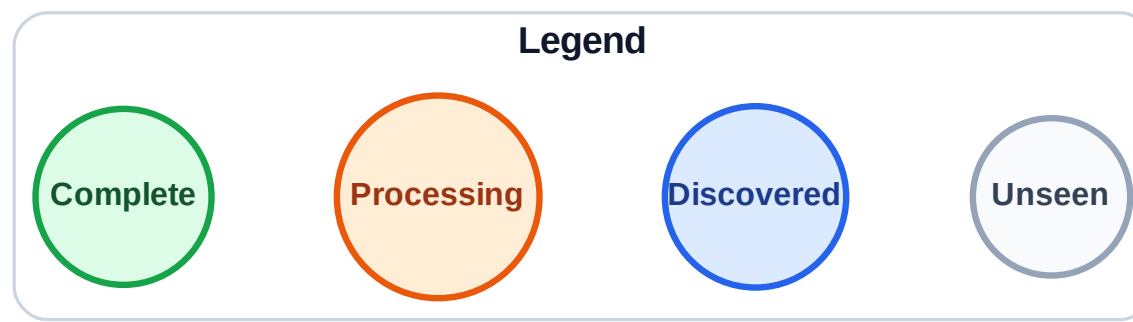
Stack (top at right): I | G | E | D | L | H | F | O

Visited: A, B, C, J



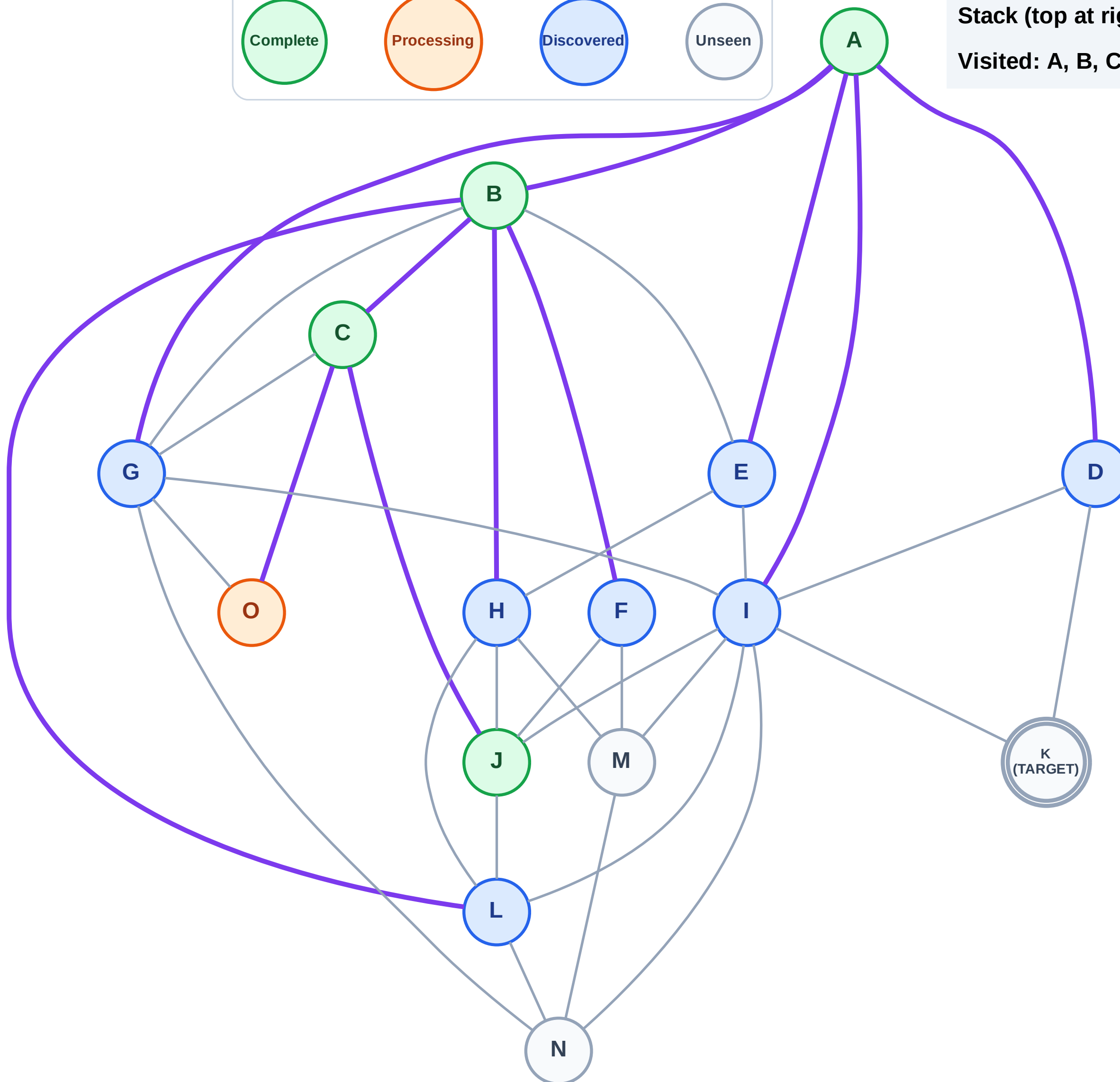
DFS Traversal

Step 10: Process node O



Stack (top at right): I | G | E | D | L | H | F

Visited: A, B, C, J



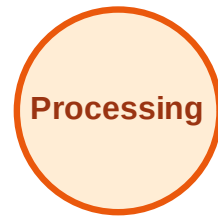
DFS Traversal

Step 11: No new neighbors from O

Legend



Complete



Processing



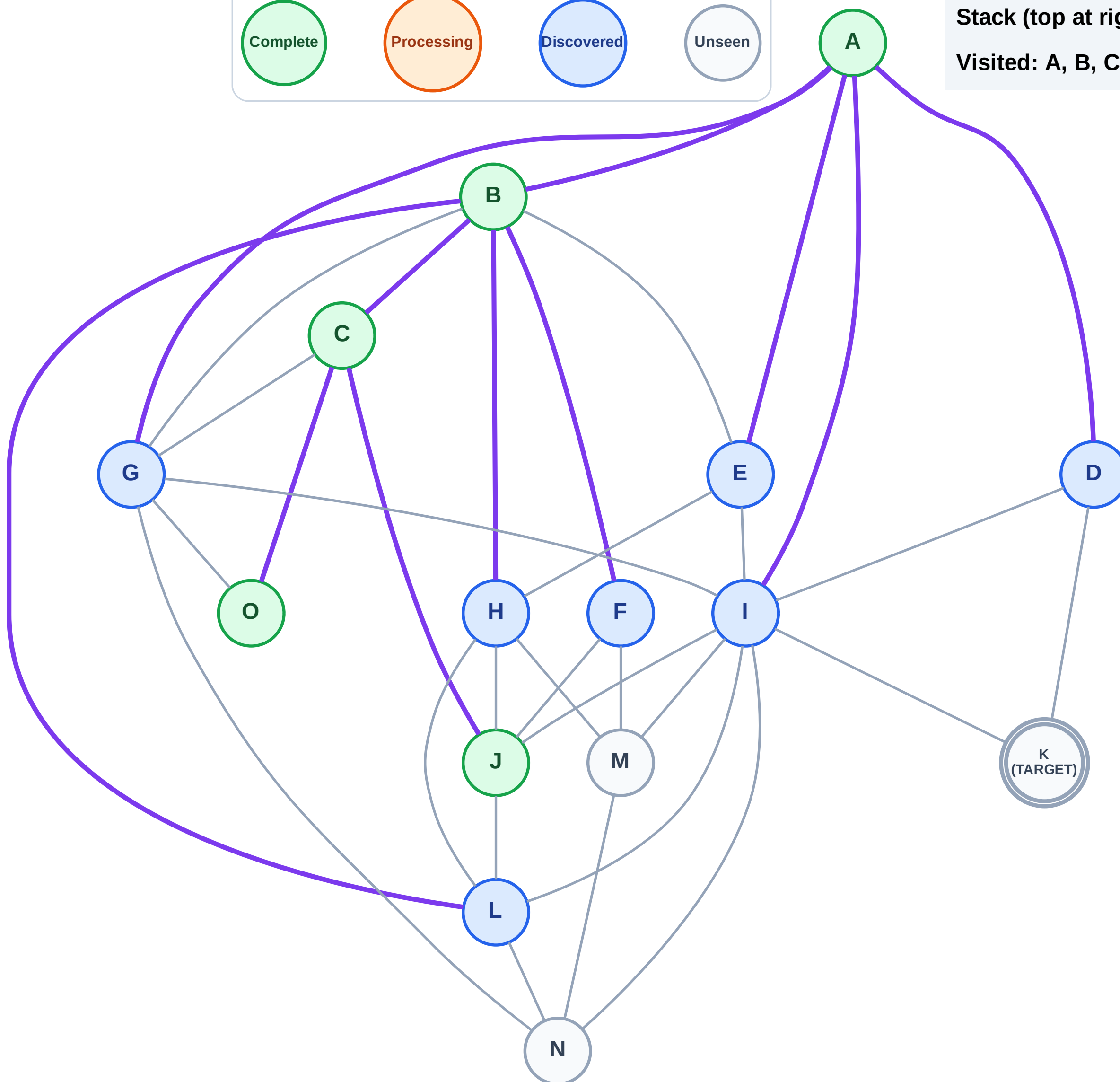
Discovered



Unseen

Stack (top at right): I | G | E | D | L | H | F

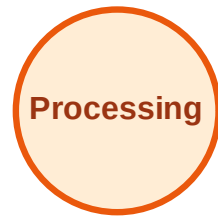
Visited: A, B, C, J, O



DFS Traversal

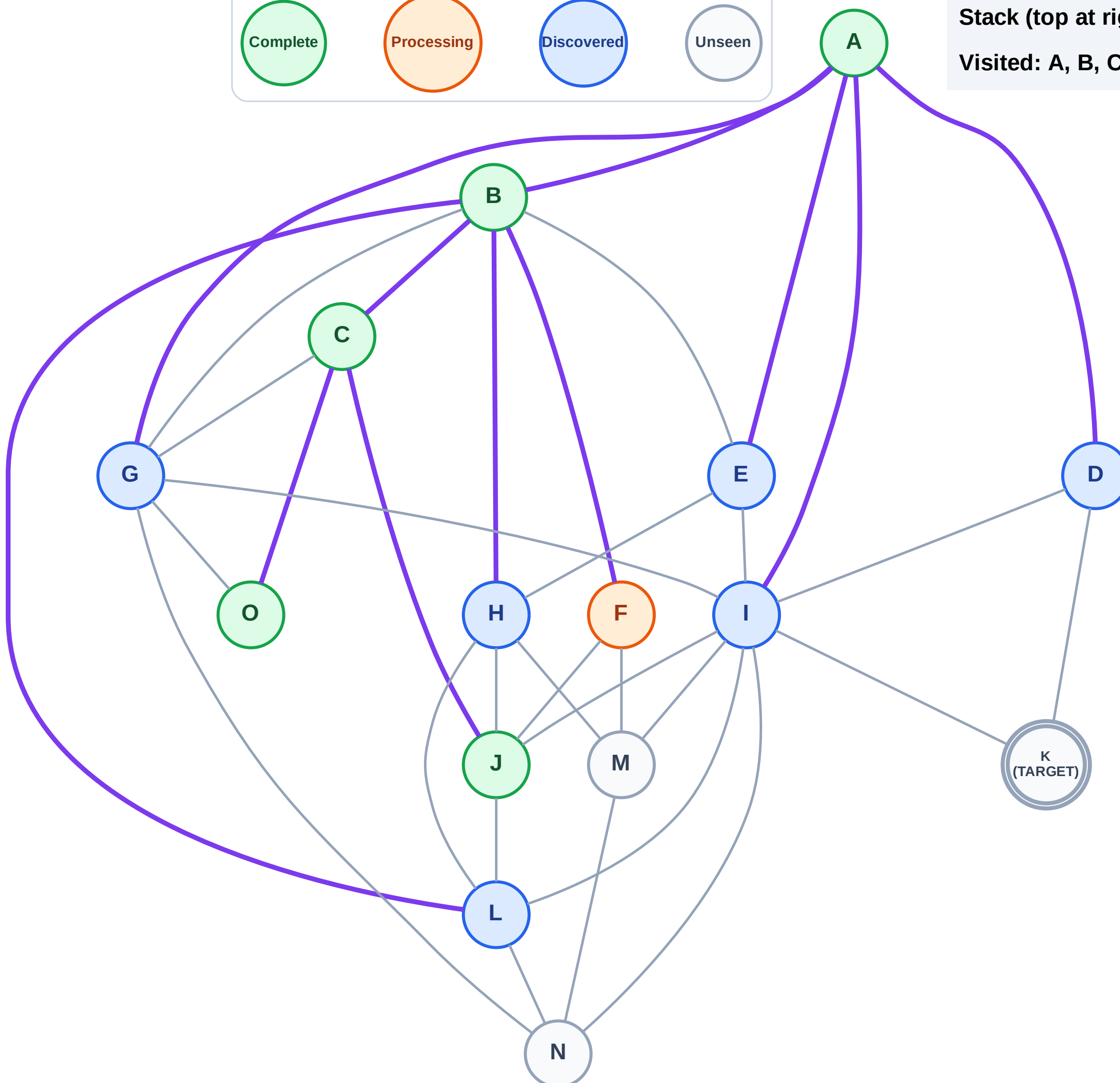
Step 12: Process node F

Legend



Stack (top at right): I | G | E | D | L | H

Visited: A, B, C, J, O



DFS Traversal

Step 13: Discover M from F

Legend



Complete



Processing



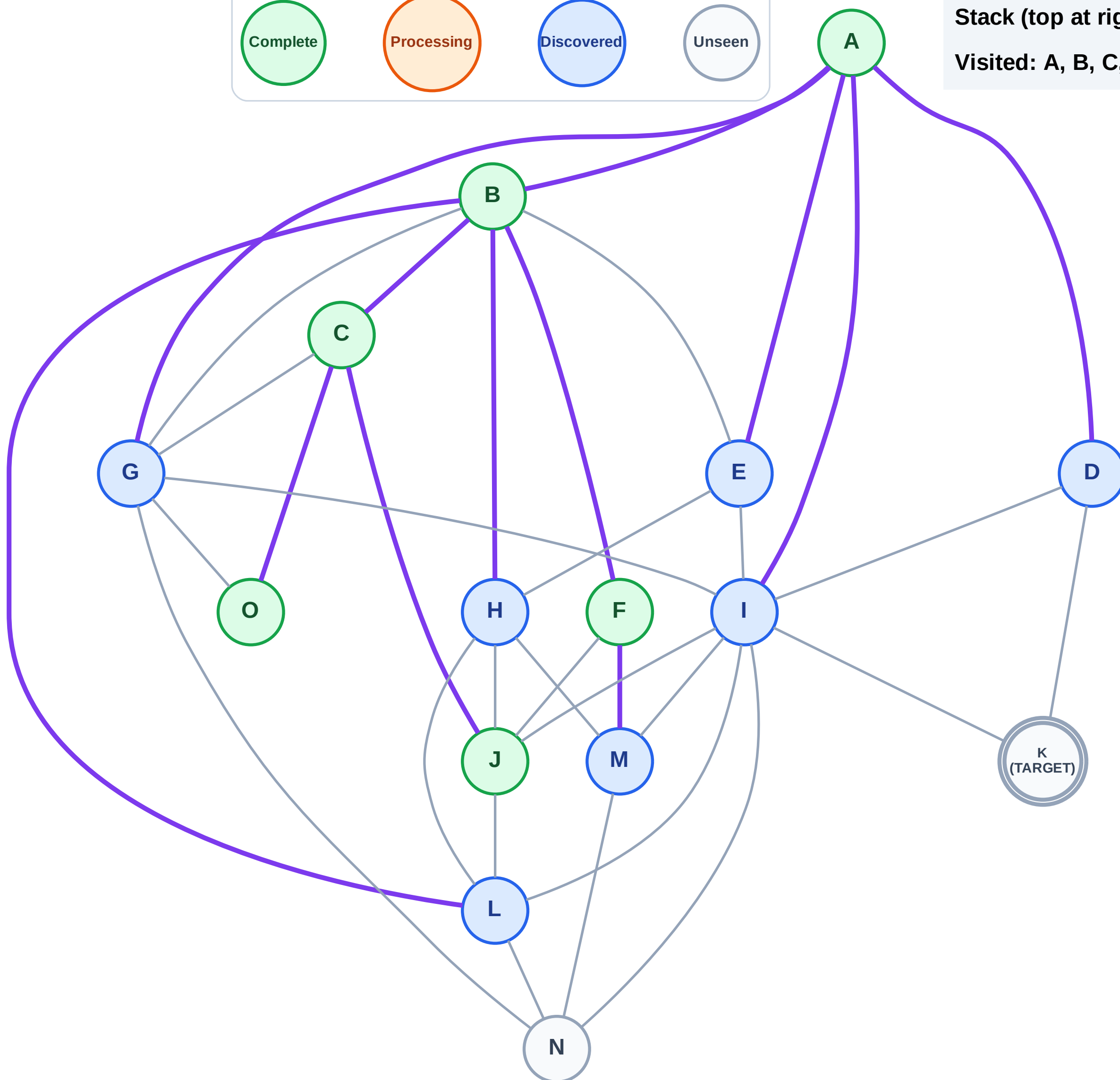
Discovered



Unseen

Stack (top at right): I | G | E | D | L | H | M

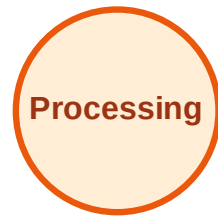
Visited: A, B, C, F, J, O



DFS Traversal

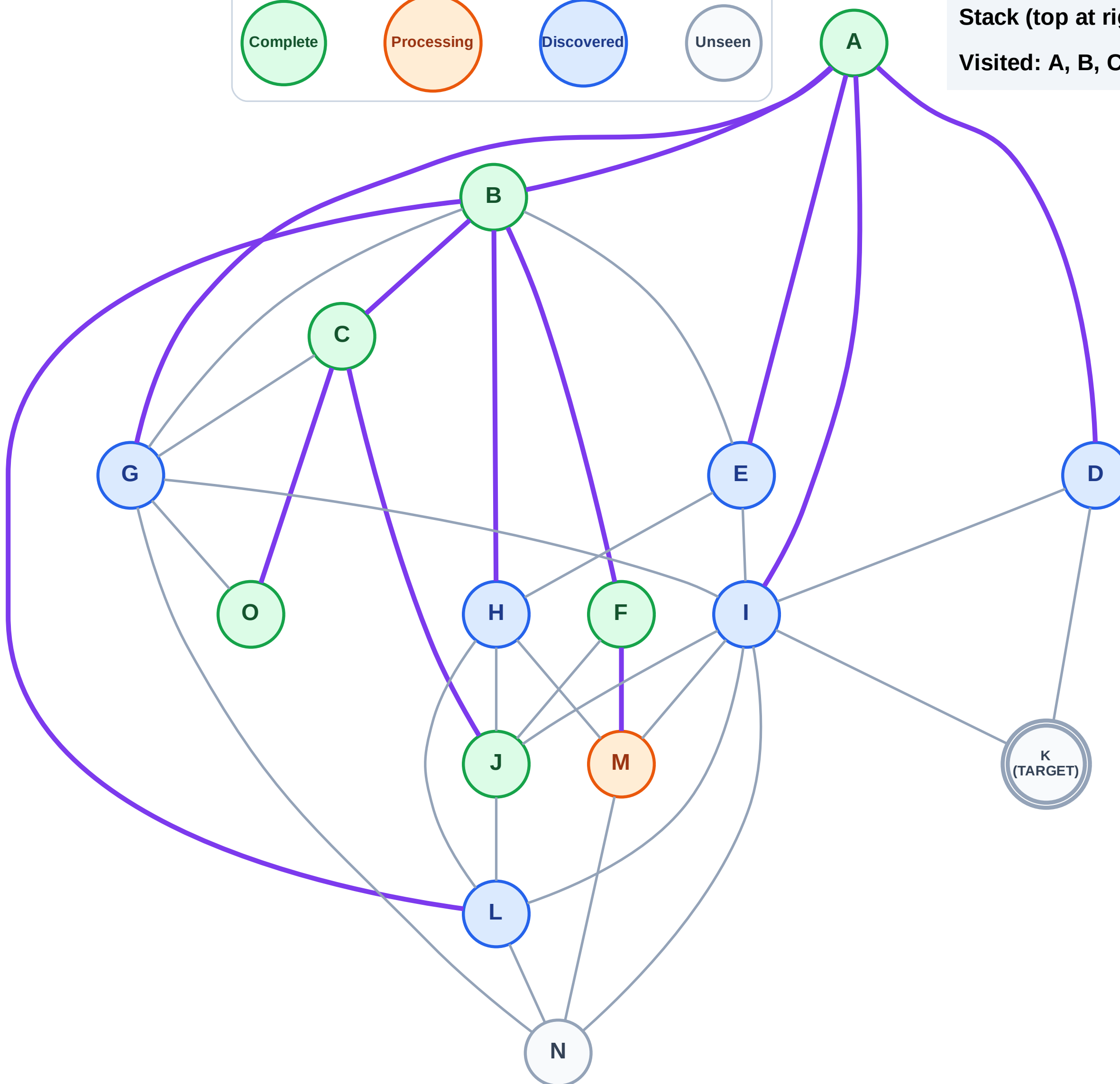
Step 14: Process node M

Legend



Stack (top at right): I | G | E | D | L | H

Visited: A, B, C, F, J, O



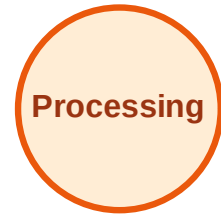
DFS Traversal

Step 15: Discover N from M

Legend



Complete



Processing



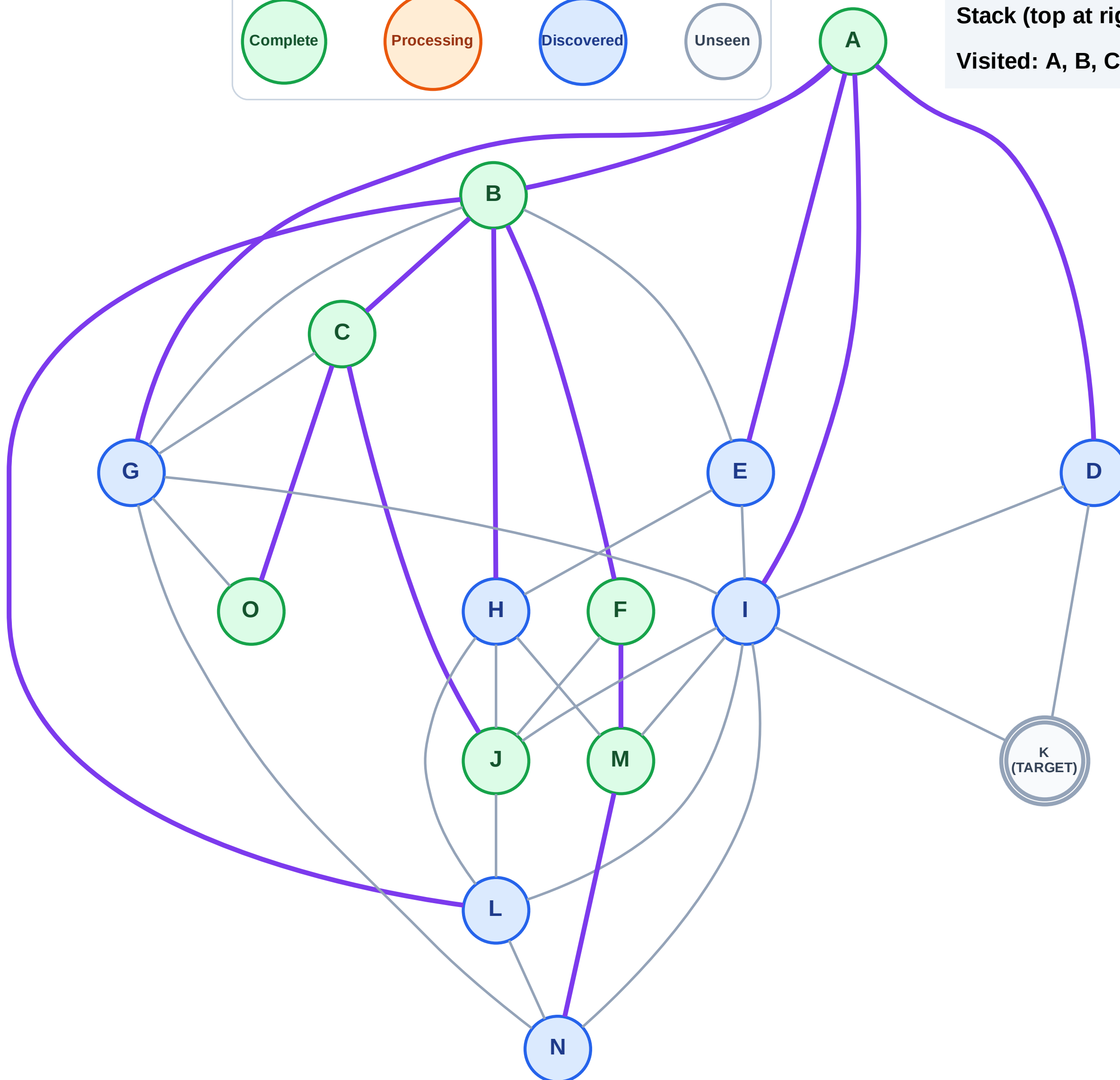
Discovered



Unseen

Stack (top at right): I | G | E | D | L | H | N

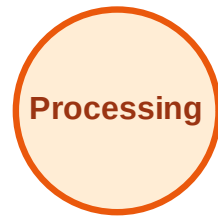
Visited: A, B, C, F, J, M, O



DFS Traversal

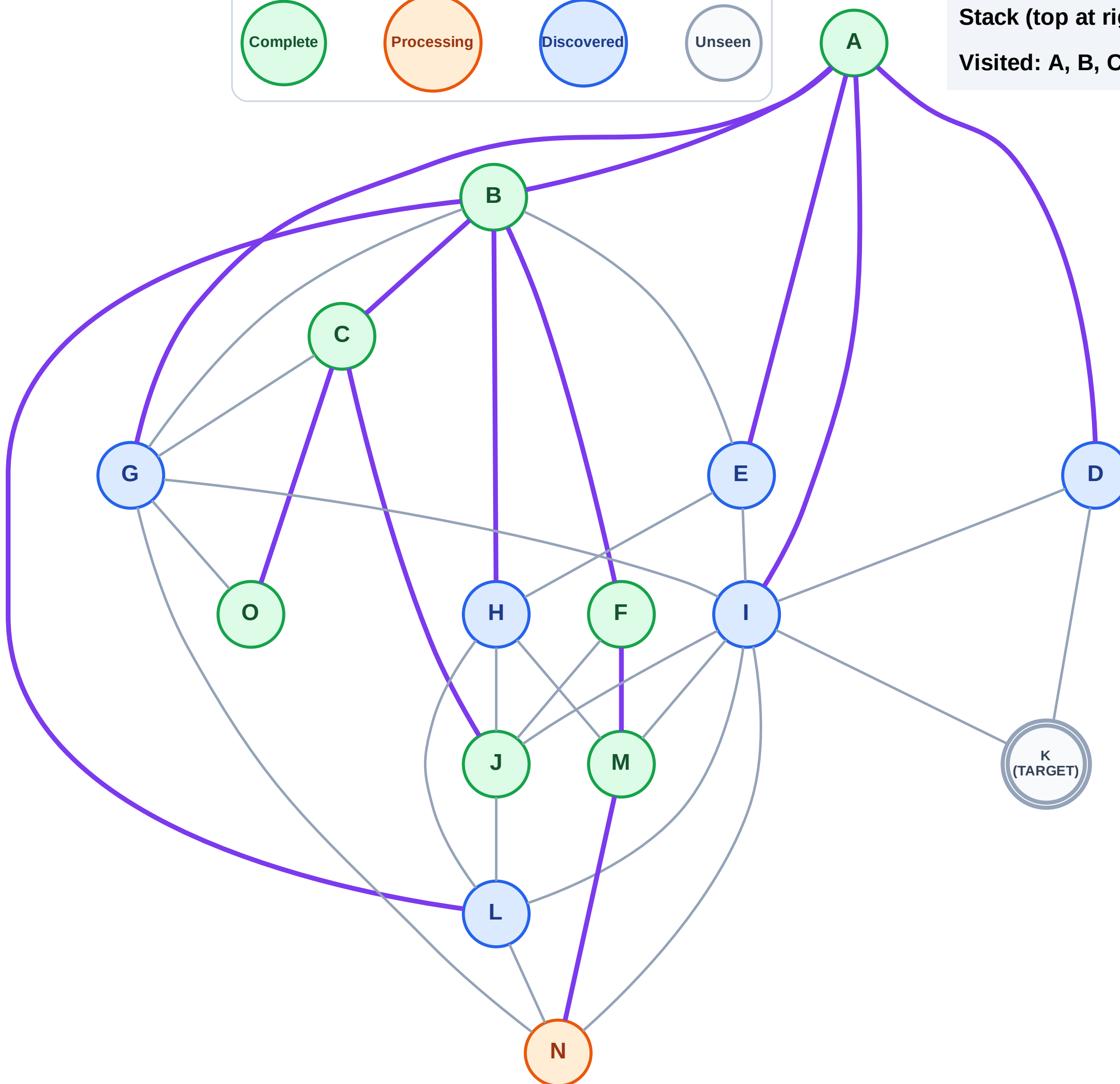
Step 16: Process node N

Legend



Stack (top at right): I | G | E | D | L | H

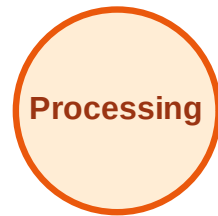
Visited: A, B, C, F, J, M, O



DFS Traversal

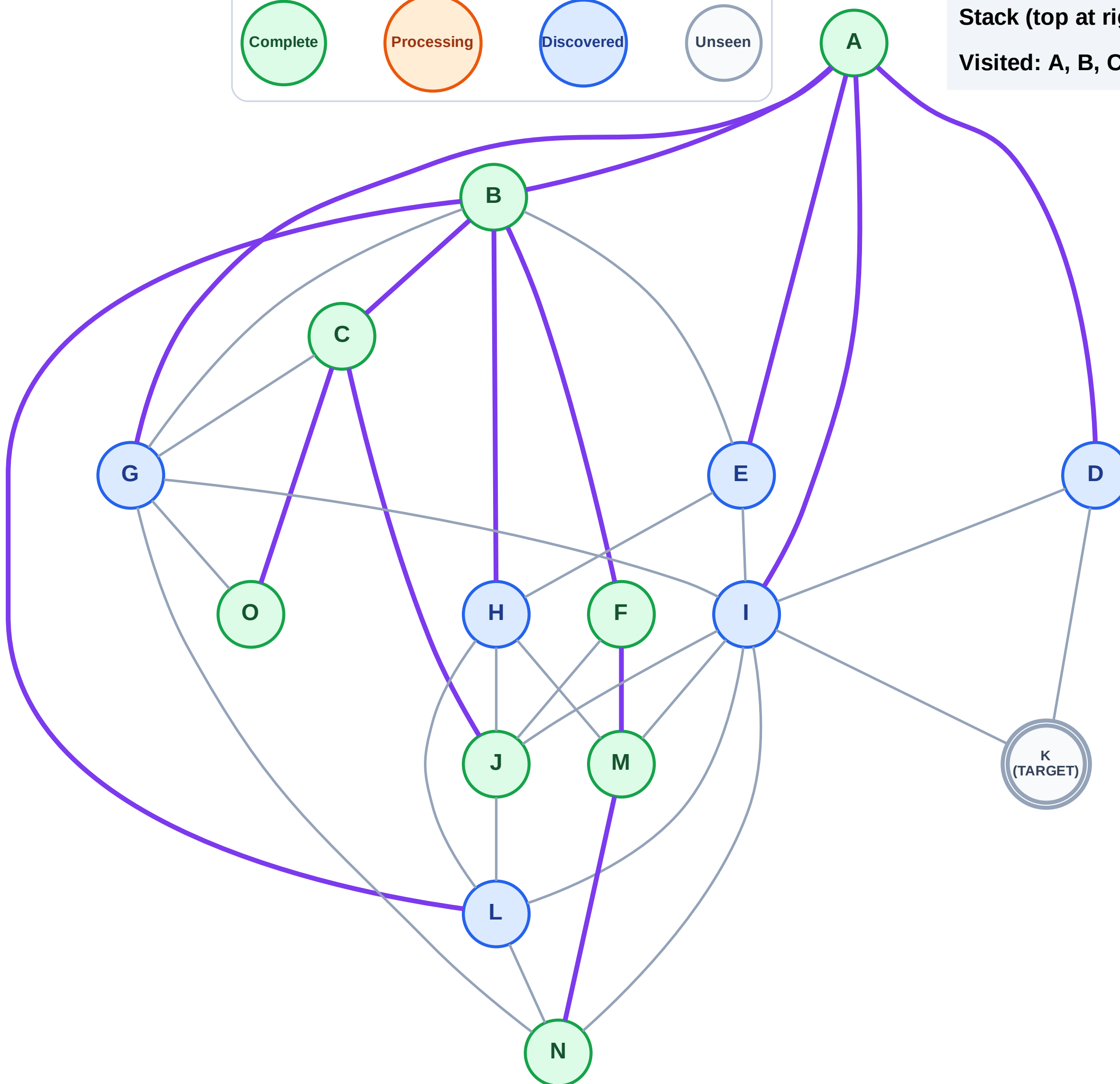
Step 17: No new neighbors from N

Legend



Stack (top at right): I | G | E | D | L | H

Visited: A, B, C, F, J, M, N, O



DFS Traversal

Step 18: Process node H

Legend



Complete



Processing



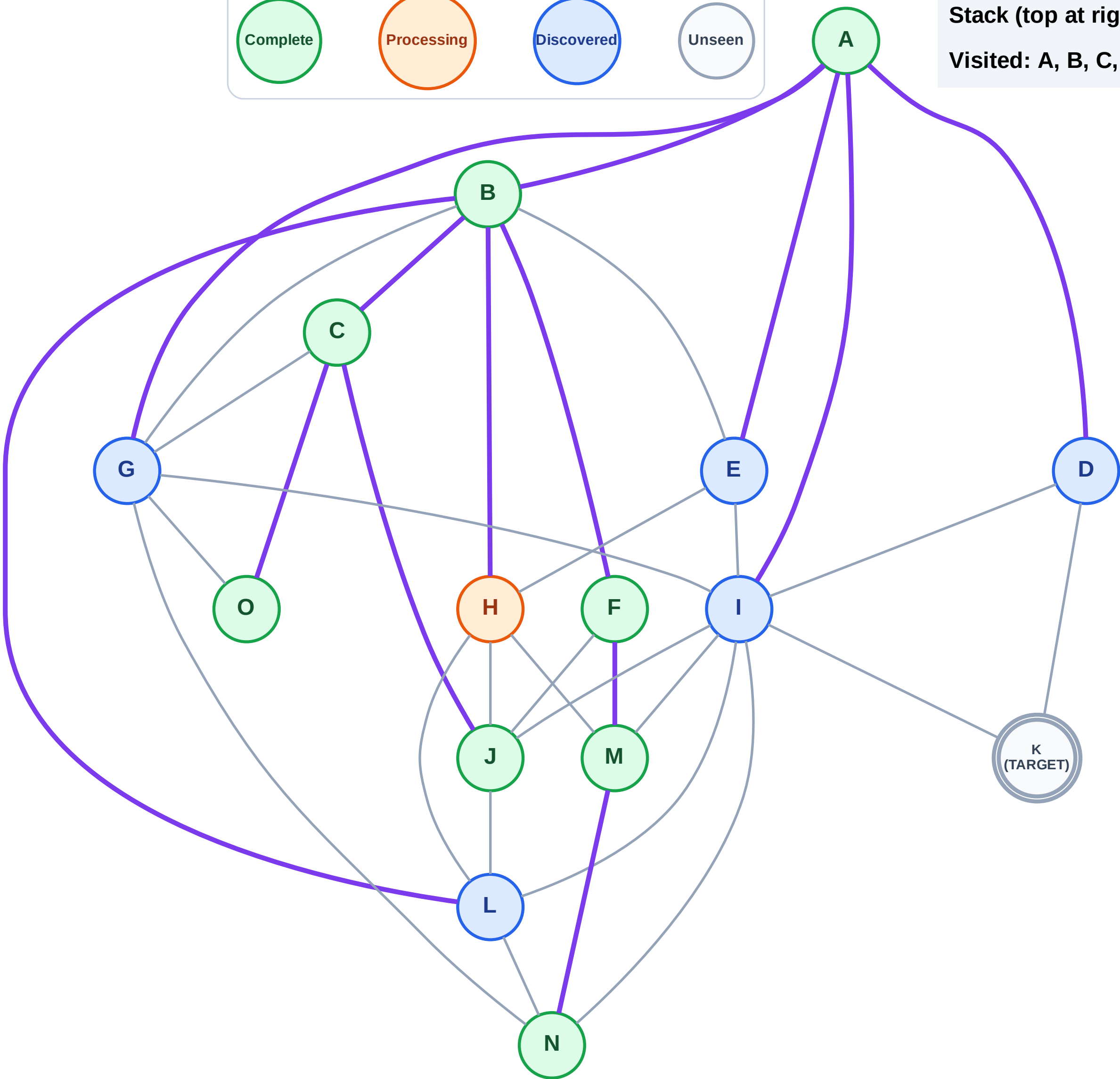
Discovered



Unseen

Stack (top at right): I | G | E | D | L

Visited: A, B, C, F, J, M, N, O



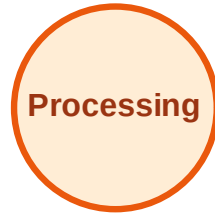
DFS Traversal

Step 19: No new neighbors from H

Legend



Complete



Processing



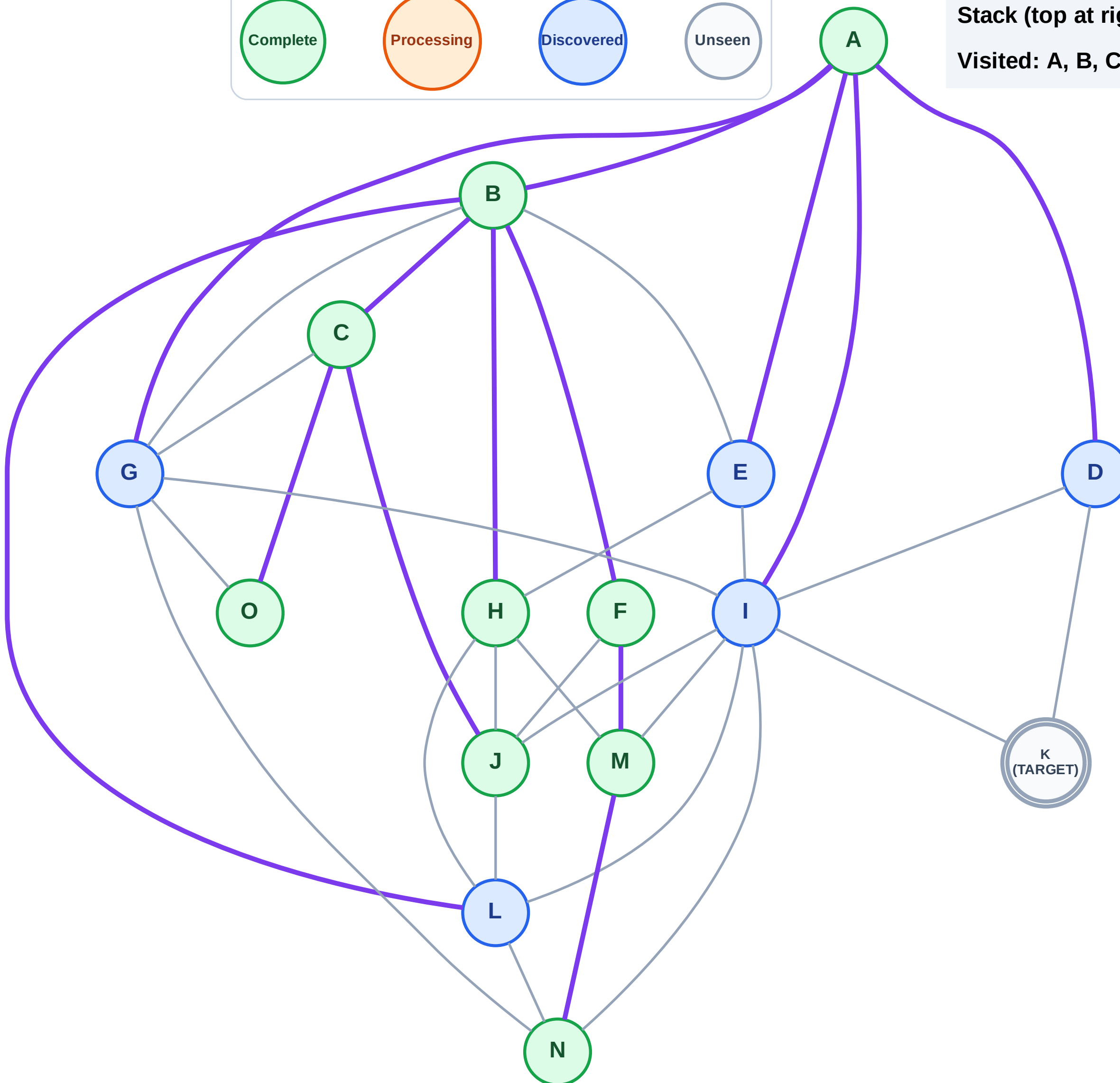
Discovered



Unseen

Stack (top at right): I | G | E | D | L

Visited: A, B, C, F, H, J, M, N, O



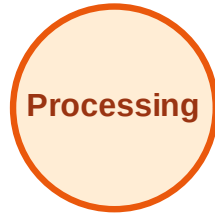
DFS Traversal

Step 20: Process node L

Legend



Complete



Processing



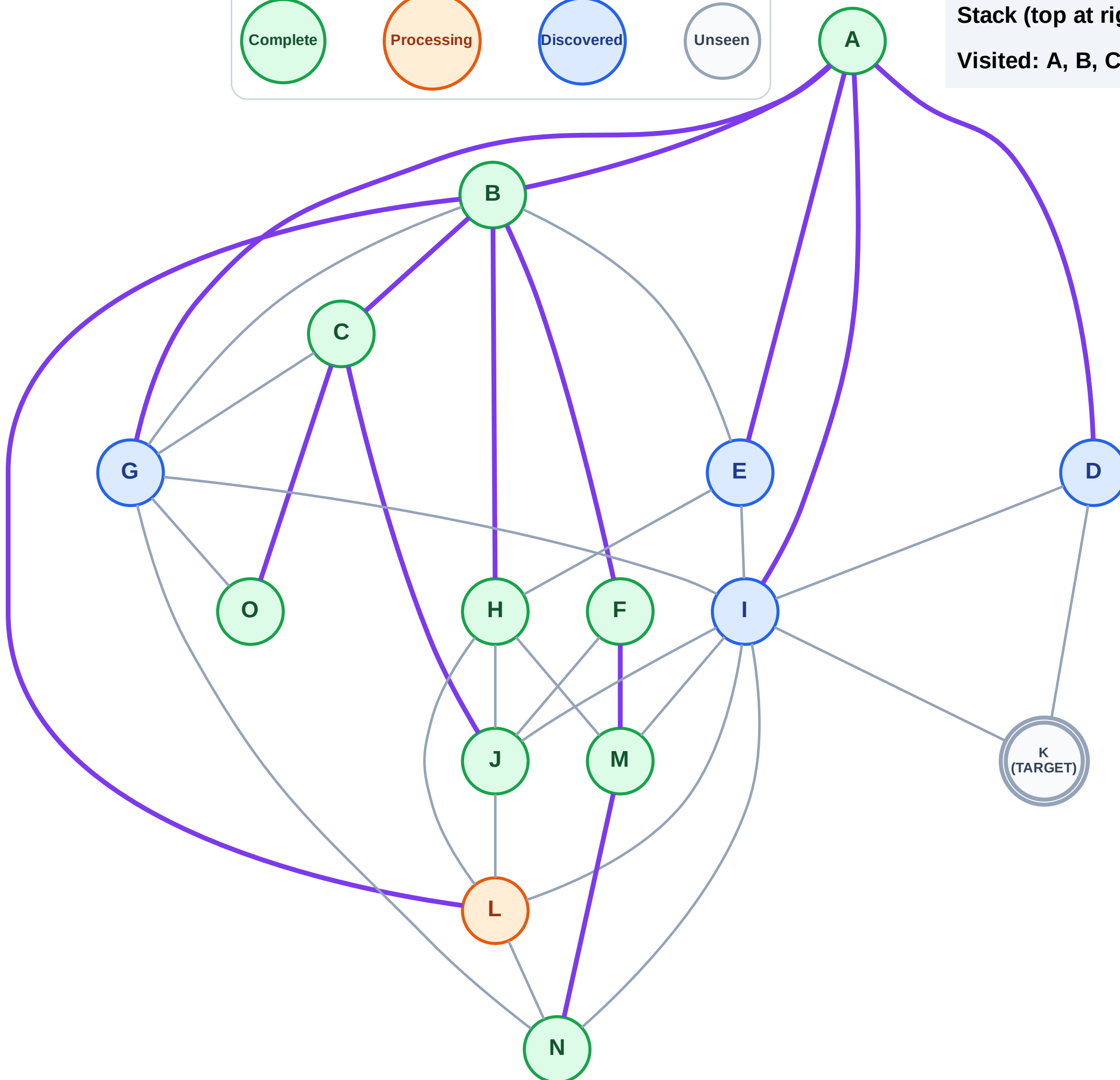
Discovered



Unseen

Stack (top at right): I | G | E | D

Visited: A, B, C, F, H, J, M, N, O



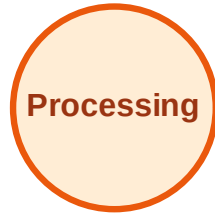
DFS Traversal

Step 21: No new neighbors from L

Legend



Complete



Processing



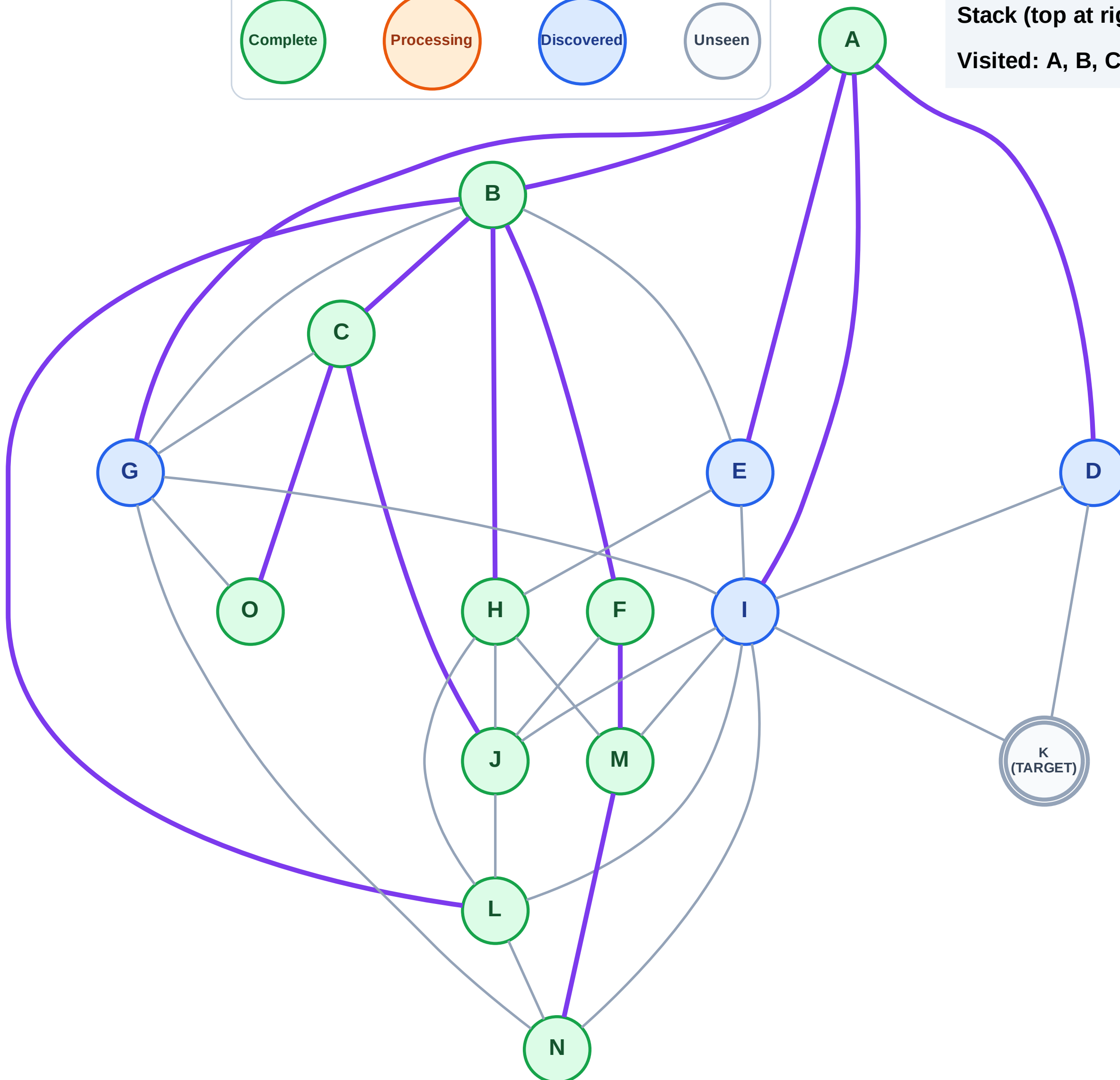
Discovered



Unseen

Stack (top at right): I | G | E | D

Visited: A, B, C, F, H, J, L, M, N, O



DFS Traversal

Step 22: Process node D

Legend



Complete



Processing



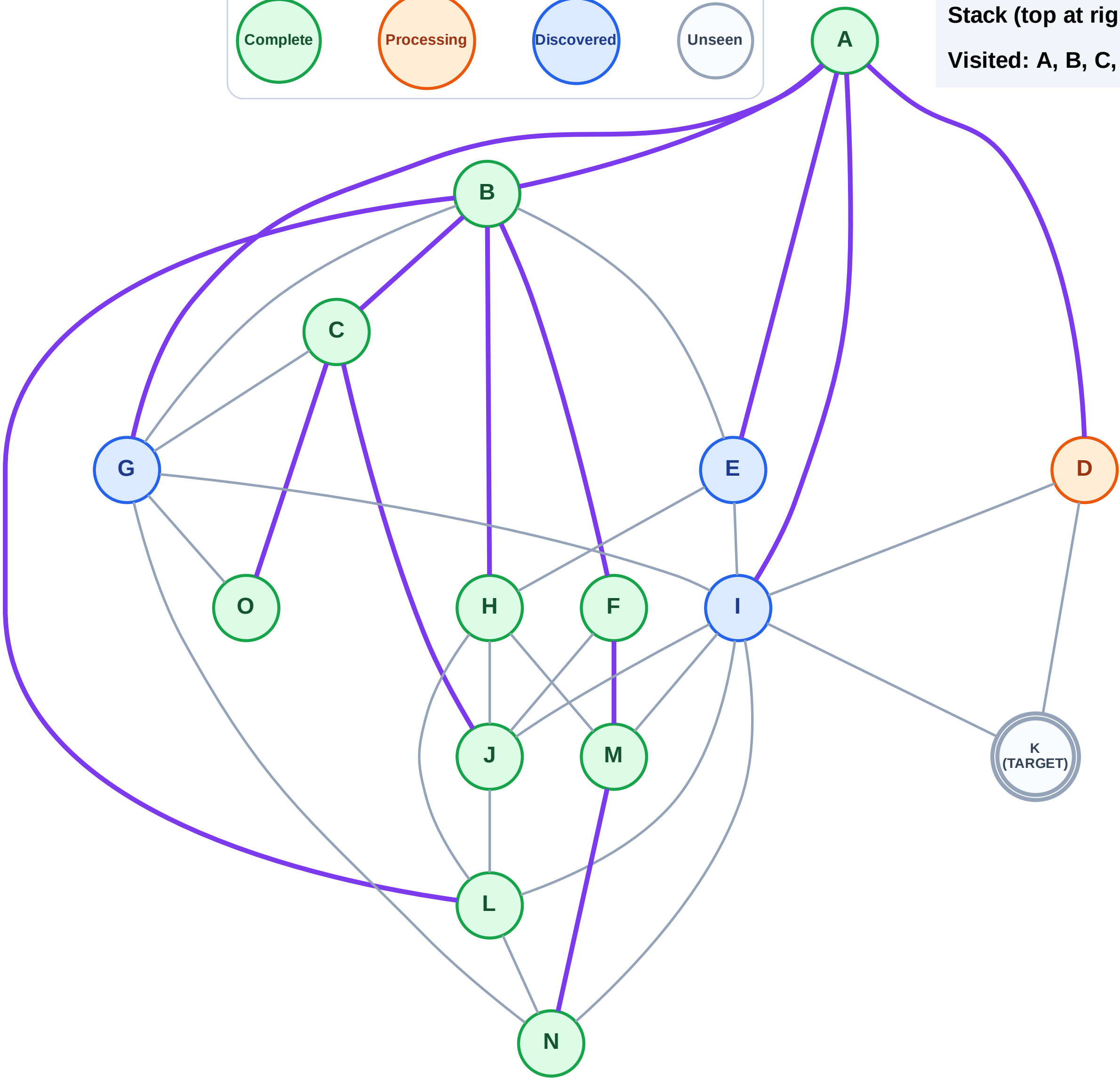
Discovered



Unseen

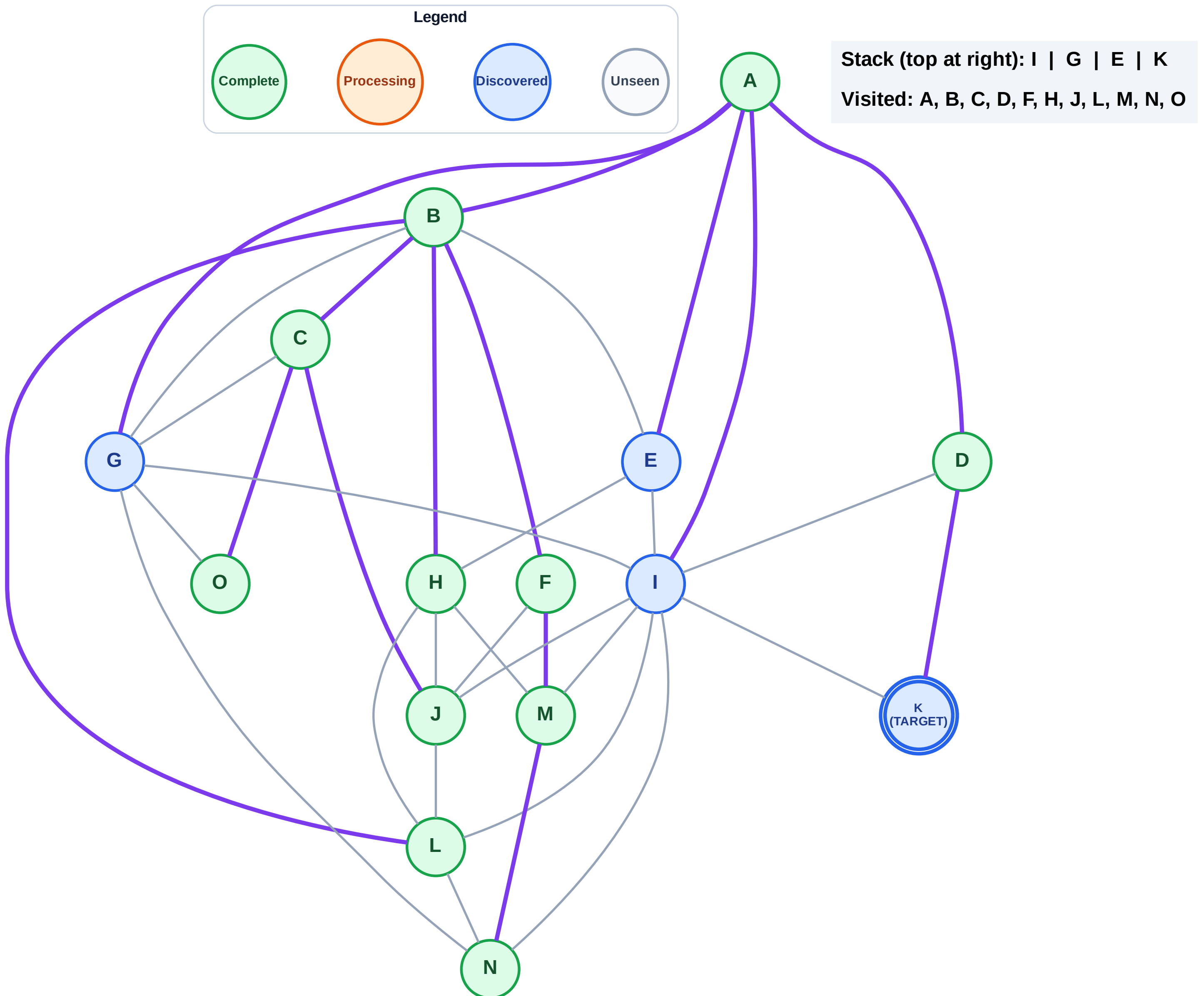
Stack (top at right): I | G | E

Visited: A, B, C, F, H, J, L, M, N, O



Step 23: Discover K from D

Step 23: Discover K from D



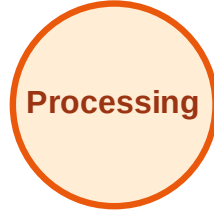
DFS Traversal

Step 24: Process node K

Legend



Complete



Processing



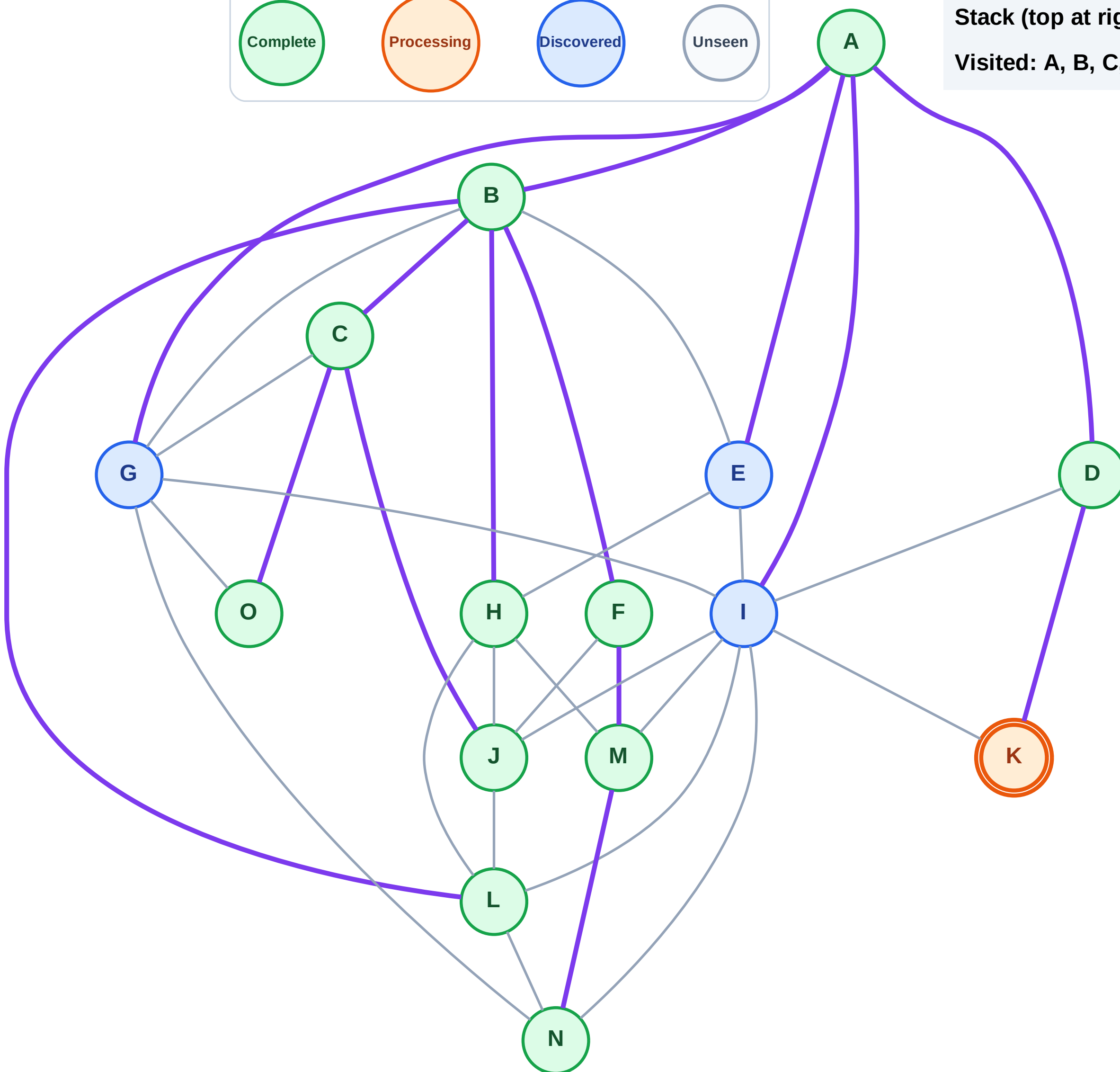
Discovered



Unseen

Stack (top at right): I | G | E

Visited: A, B, C, D, F, H, J, L, M, N, O



DFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): I | G | E
Visited: A, B, C, D, F, H, J, K, L, M, N, O

